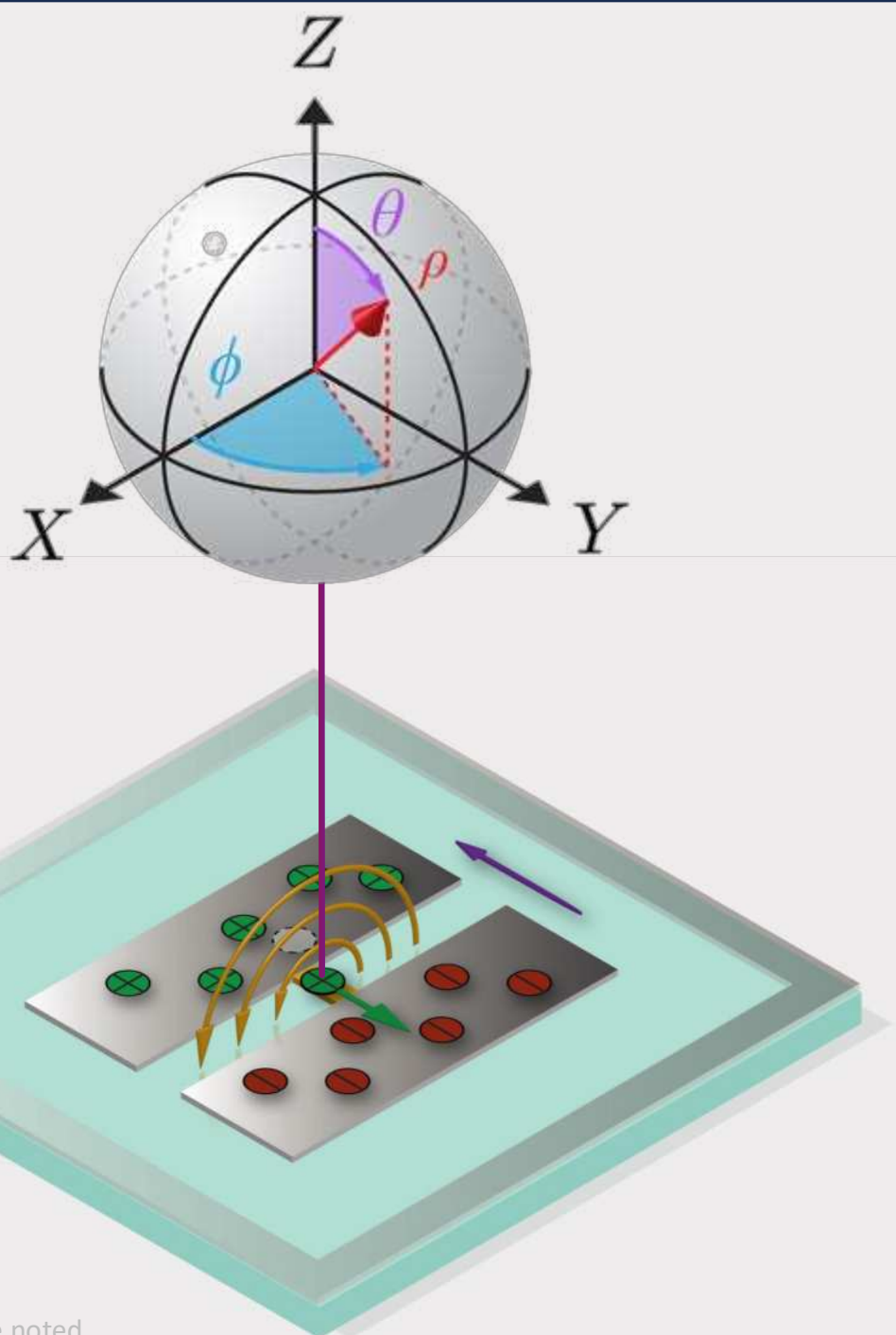




Your First Superconducting Qubit

Introduction to **Circuit** Quantum Electrodynamics (cQED)

Zlatko K. Mineev

Current Affiliation:

Google Quantum AI; Assoc. Fellow @ CIFAR

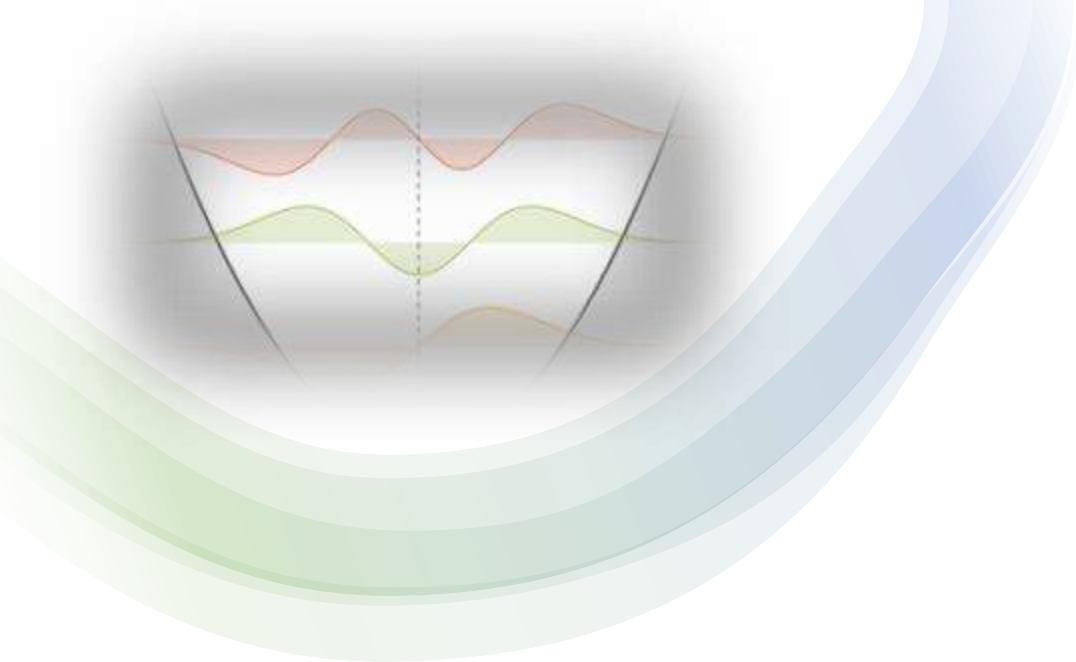
Former Affiliation: IBM Quantum; Yale University



@zlatko_mineev



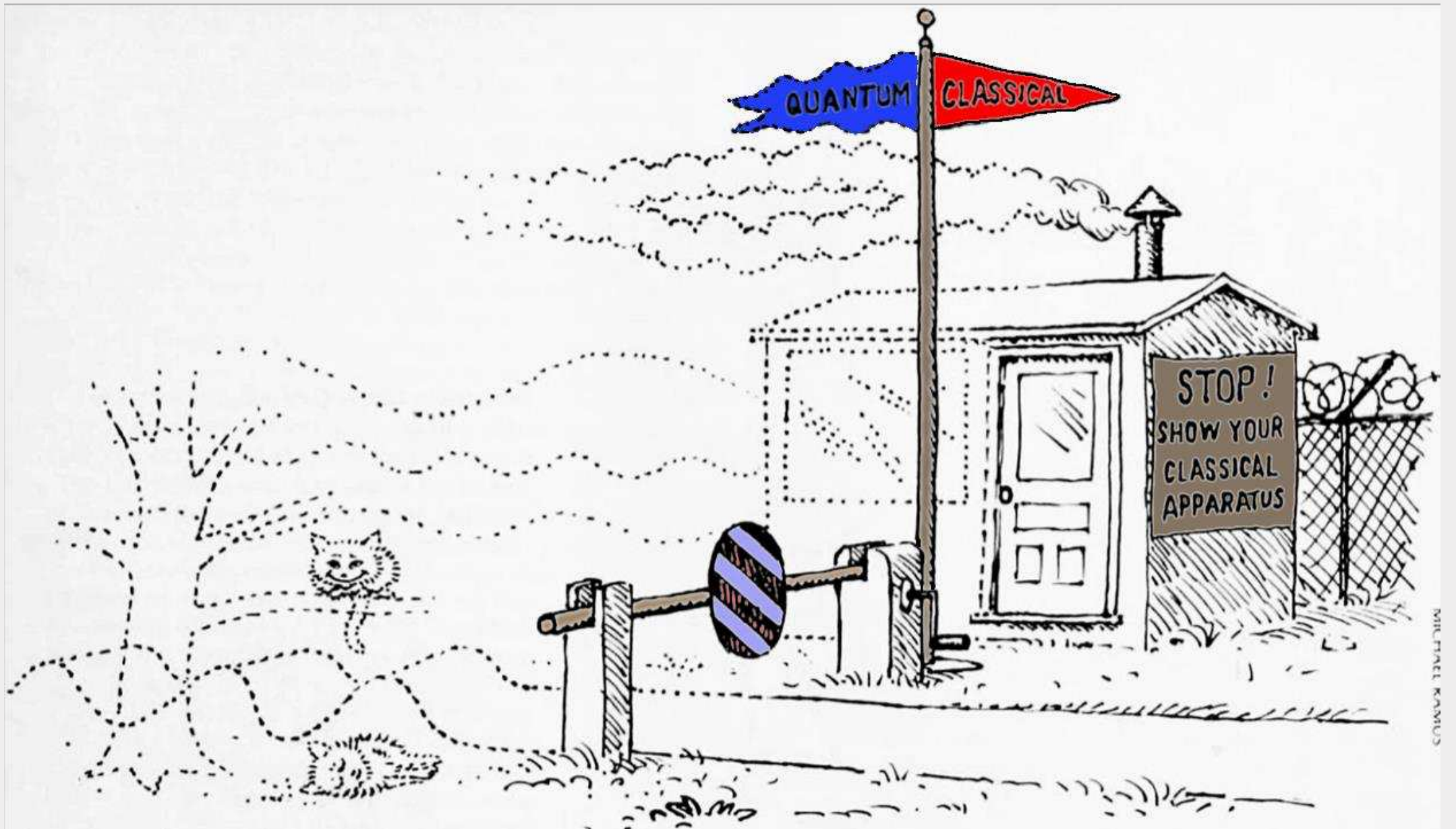
zlatko-mineev.com/blog



Unveiling the quantum

Quantum harmonic oscillator





Drawing: Zurek, Physics Today (1991)



Dirac's canonical quantization



Source: Cambridge University,
Cavendish Laboratory / Wikimedia
Commons

Quantum (commutator)

$$-\frac{i}{\hbar} [\hat{x}, \hat{p}] = \hat{1}$$

$$\frac{d}{dt} \hat{O} = -\frac{i}{\hbar} [\hat{O}, \hat{H}]$$

where $\hat{O}(\hat{x}, \hat{p}; t), \hat{H}(\hat{x}, \hat{p}; t)$

These look
a lot alike!

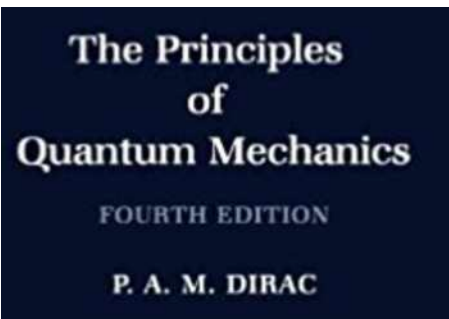
Classical (Poisson bracket)

$$\{x, p\}_P = 1$$

$$\frac{d}{dt} O = \{O, \mathcal{H}\}_P$$

where $O(x, p; t), \mathcal{H}(x, p; t)$

Procedure: Supplant the Poisson brackets by commutators



$$\{x, p\} \longmapsto \frac{1}{i\hbar} [\hat{x}, \hat{p}]$$

$$\{O, O'\} \longmapsto \frac{1}{i\hbar} [\hat{O}, \hat{O}']$$

“There is, however, a fairly general method of obtaining quantum conditions, applicable to a very large class of dynamical systems. This is the **method of classical analogy**”

P.A.M. Dirac

If I knew what I was doing, it
wouldn't be called research.

Albert Einstein
See Hawken *et al.* (2010)

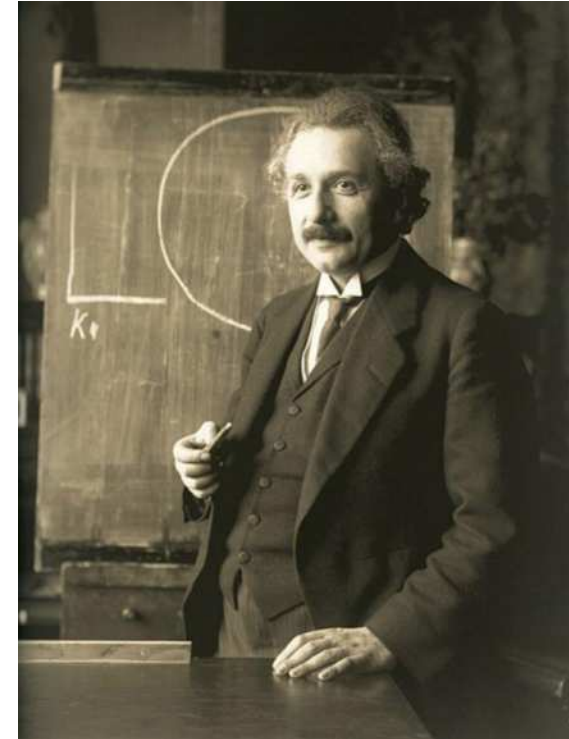
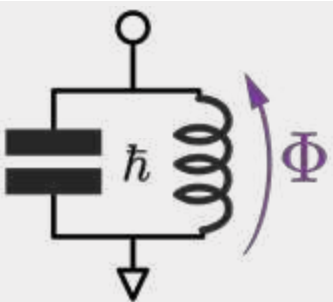


Photo: F. Schmutzer



The classical and quantum oscillator

Classical

Quantum

$$\Phi(t) \mapsto \hat{\Phi}$$

$$Q(t) \mapsto \hat{Q}$$

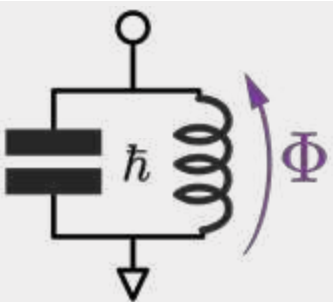
$$\mathcal{H}(\Phi, Q) \mapsto \hat{H}(\hat{\Phi}, \hat{Q})$$

$$\{\Phi, Q\} = 1 \mapsto [\hat{\Phi}, \hat{Q}] = i\hbar \hat{1}$$

$$\{\alpha, \alpha^*\} = 1/(i\hbar) \mapsto [\hat{a}, \hat{a}^\dagger] = \hat{1}$$

$$\{A, B\} = \frac{\partial A}{\partial \Phi} \frac{\partial B}{\partial Q} - \frac{\partial B}{\partial \Phi} \frac{\partial A}{\partial Q}$$

$$[\hat{A}, \hat{B}] = \hat{A}\hat{B} - \hat{B}\hat{A}$$



The classical and quantum oscillator



Classical

Quantum

Hamiltonian

$$\begin{aligned}\mathcal{H} &= \frac{Q^2}{2C} + \frac{\Phi^2}{2L} \\ &= \frac{1}{2} \hbar \omega_0 (\alpha^* \alpha + \alpha \alpha^*)\end{aligned}$$

$$\begin{aligned}\hat{H} &= \frac{\hat{\Phi}^2}{2L} + \frac{\hat{Q}^2}{2C} \\ &= \hbar \omega_0 (\hat{a}^\dagger \hat{a} + \hat{a} \hat{a}^\dagger)\end{aligned}$$

Phase space

$$\alpha(t) = \sqrt{\frac{1}{2\hbar Z}} [\Phi(t) + iZQ(t)]$$

$$\alpha(t) = \alpha(0) e^{-i\omega_0 t}$$

$$\Phi(t) = \sqrt{\frac{\hbar Z}{2}} (\alpha^*(t) + \alpha(t))$$

$$Q(t) = i\sqrt{\frac{\hbar}{2Z}} (\alpha^*(t) - \alpha(t))$$

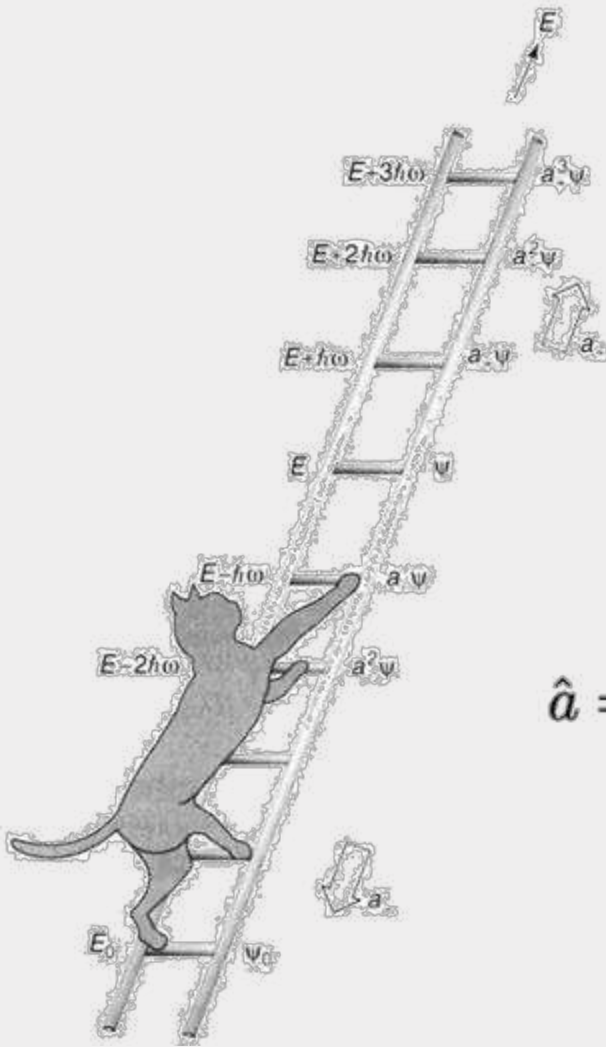
$$\hat{a} = \sqrt{\frac{1}{2\hbar Z}} (\hat{\Phi} + iZ\hat{Q})$$

$$\hat{a}(t) = \hat{a}(0) e^{-i\omega_0 t} \quad (\text{Heisenberg picture})$$

$$\hat{\Phi} = \Phi_{\text{ZPF}} (\hat{a}^\dagger + \hat{a})$$

$$\hat{Q} = iQ_{\text{ZPF}} (\hat{a}^\dagger - \hat{a})$$

Ladder operators and matrix representation



annihilation

$$\hat{a} |0\rangle = 0$$

$$\hat{a} |1\rangle = \sqrt{1} |0\rangle$$

creation

$$\hat{a}^\dagger |0\rangle = \sqrt{1} |1\rangle$$

$$\hat{a}^\dagger |1\rangle = \sqrt{2} |2\rangle$$

general hopping

$$\hat{a} |n\rangle = \sqrt{n} |n-1\rangle$$

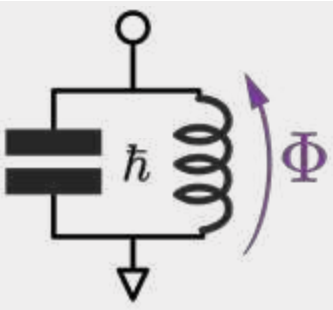
$$\hat{a}^\dagger |n\rangle = \sqrt{n+1} |n+1\rangle$$

$$\hat{a} = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & \sqrt{2} & 0 \\ 0 & 0 & 0 & \sqrt{3} \\ 0 & 0 & 0 & \ddots \end{pmatrix}$$

$$\hat{a}^\dagger = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & \sqrt{2} & 0 & 0 \\ 0 & 0 & \sqrt{3} & \ddots \end{pmatrix}$$

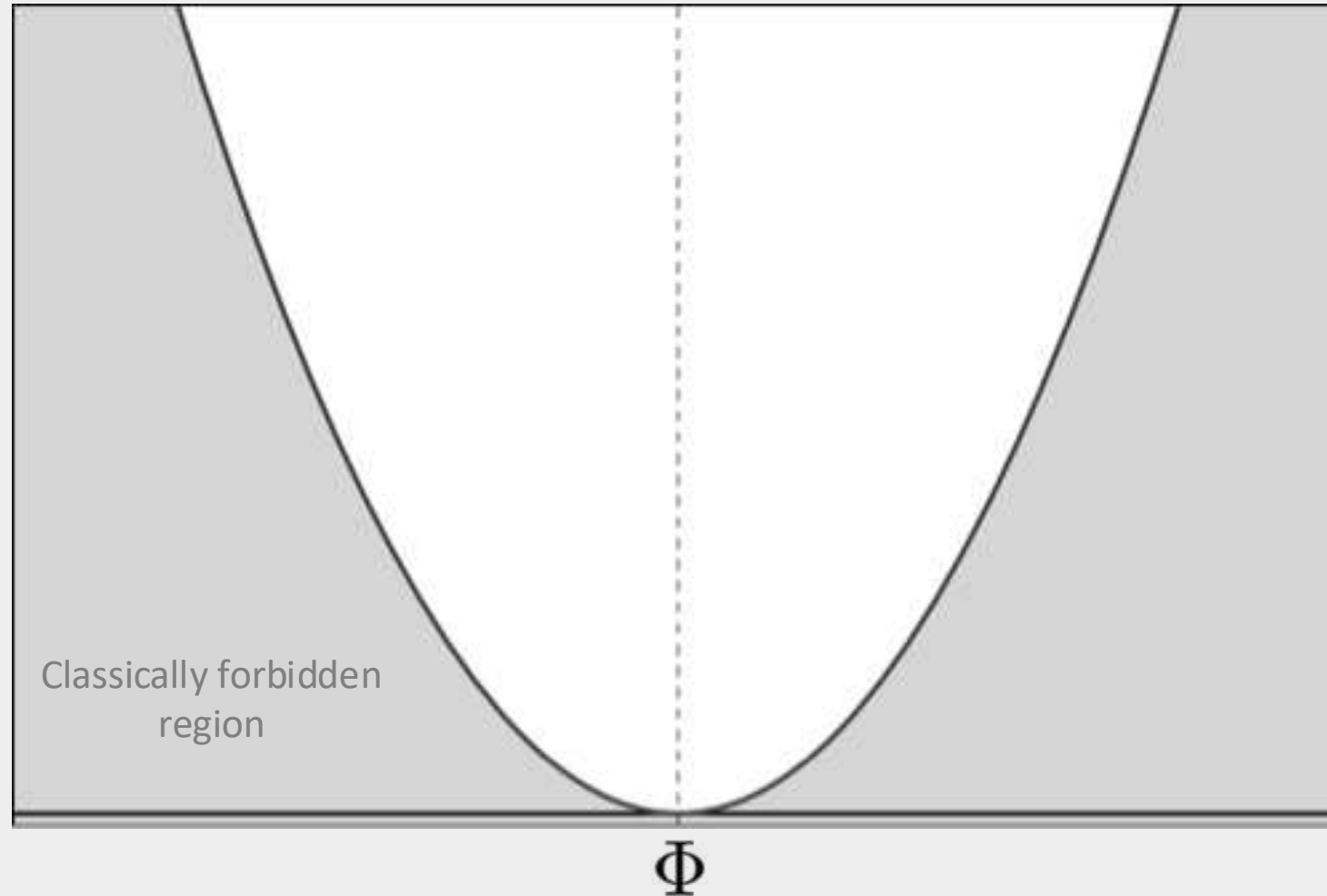
$$\hat{a}^\dagger \hat{a} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & \ddots \end{pmatrix}$$

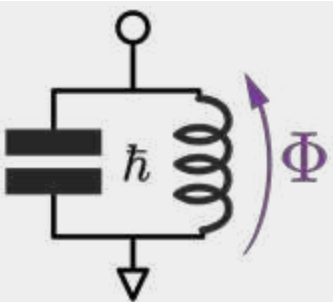
Wavefunctions of the quantum oscillator



$$\mathcal{E}_{\text{ind}}(\Phi) = \frac{\Phi^2}{2L}$$

Energy



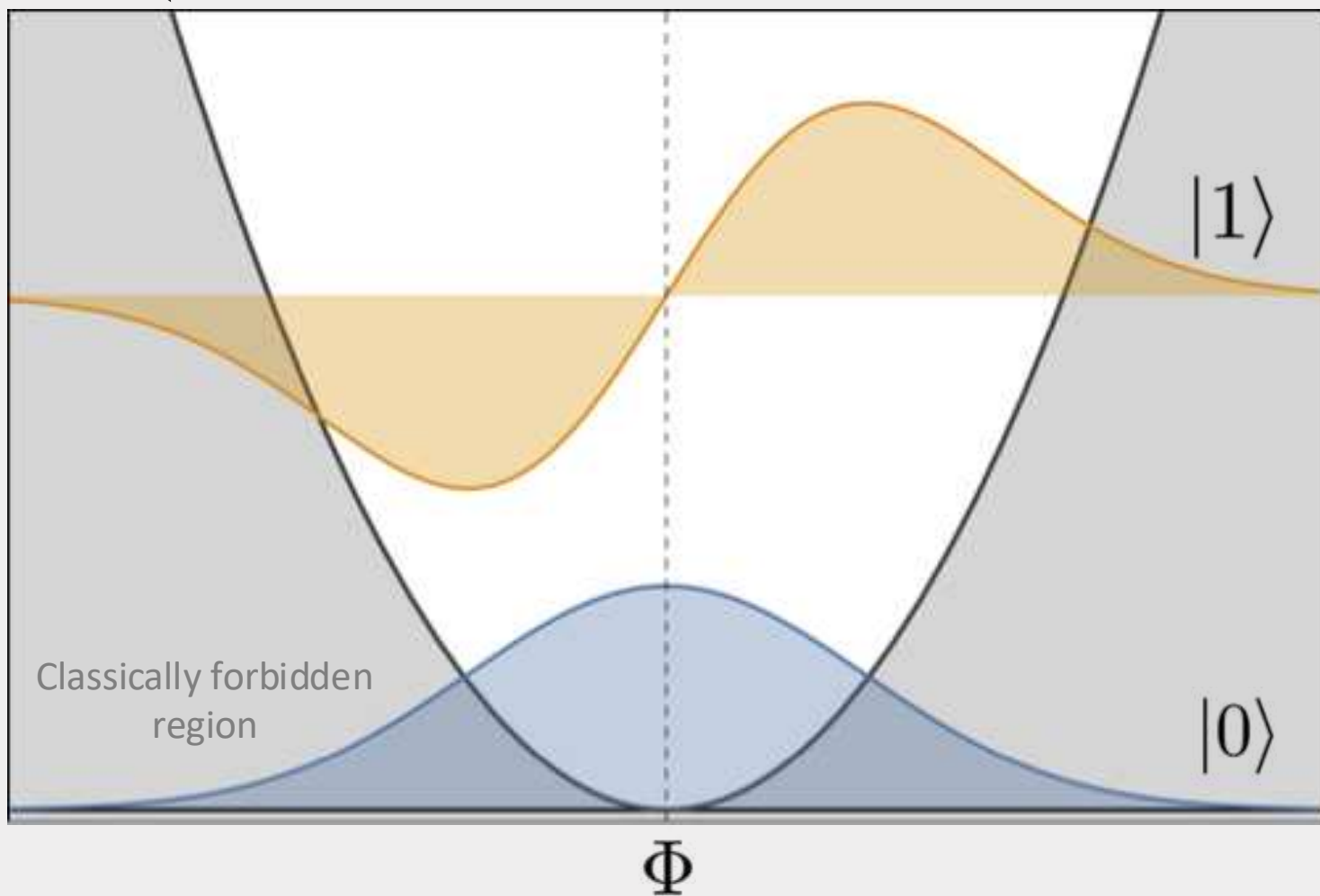


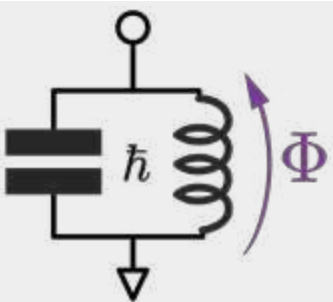
Wavefunctions of the quantum oscillator

$$\mathcal{E}_{\text{ind}}(\Phi) = \frac{\Phi^2}{2L}$$

$$\psi_n(\Phi) \equiv \langle \Phi | n \rangle$$

Energy /
Scaled
wavefunction
amplitude



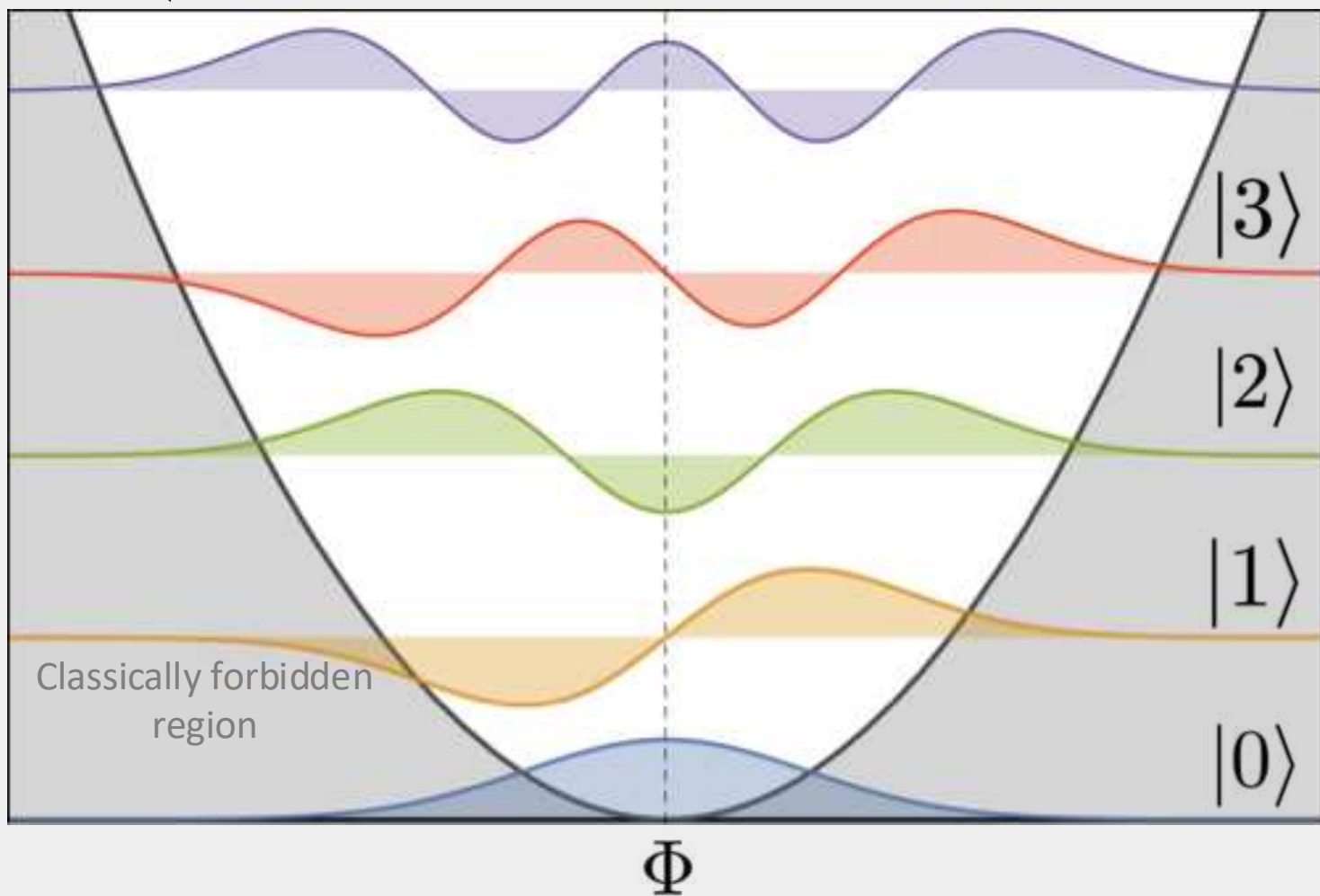


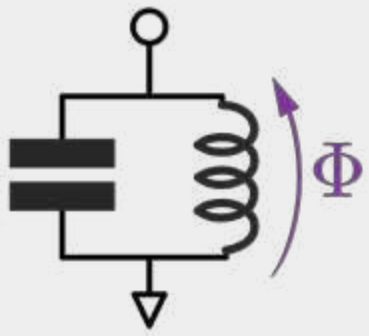
Wavefunctions of the quantum oscillator

$$\mathcal{E}_{\text{ind}}(\Phi) = \frac{\Phi^2}{2L}$$

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Energy /
Scaled
wavefunction
amplitude



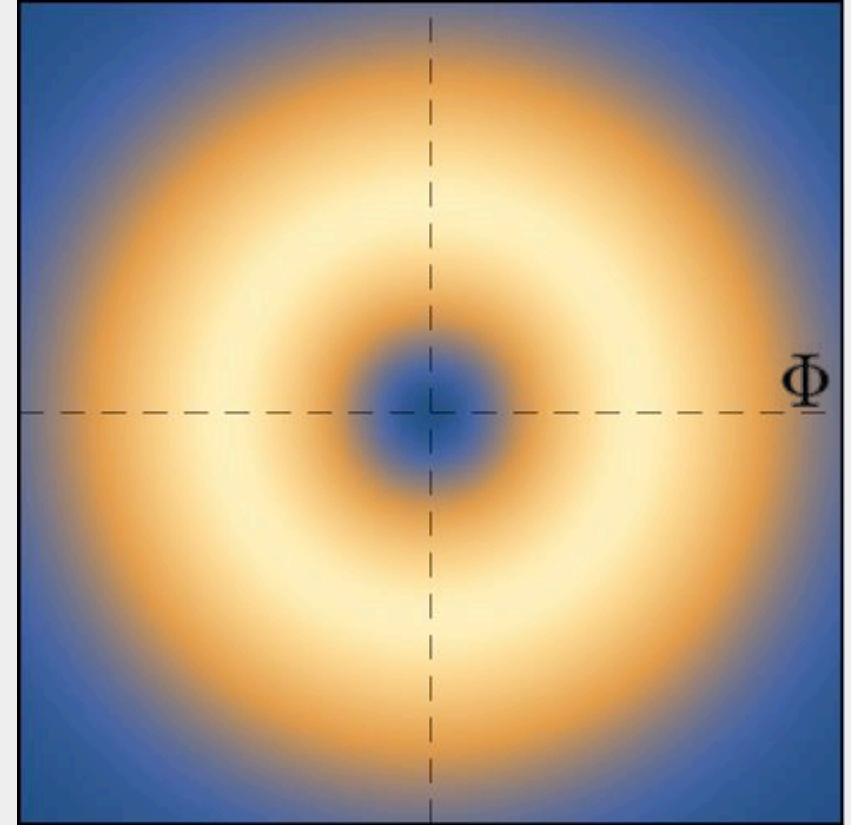
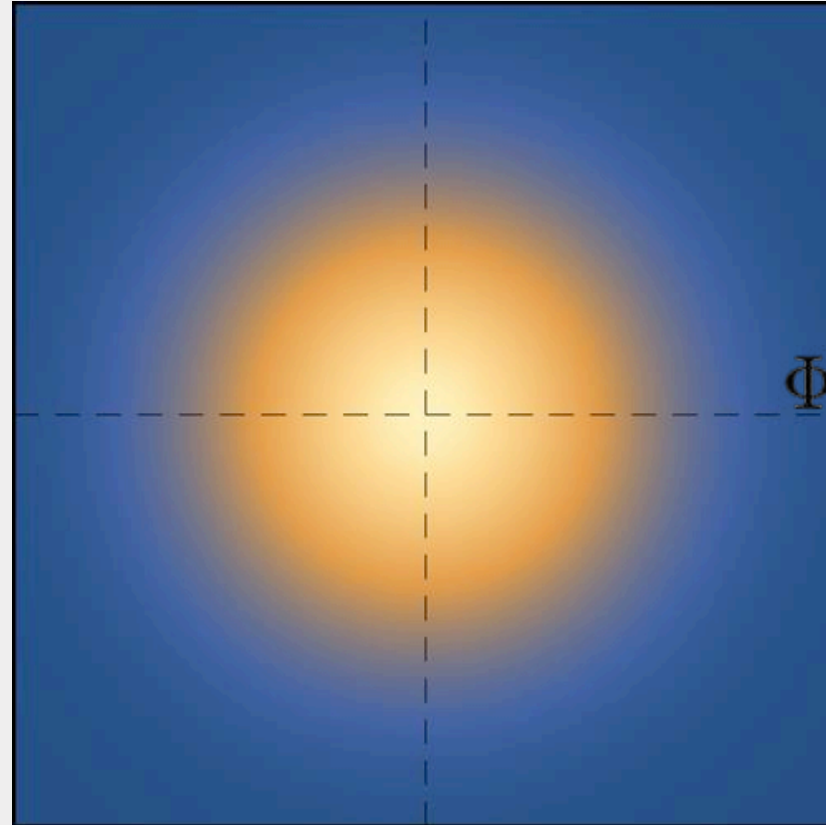
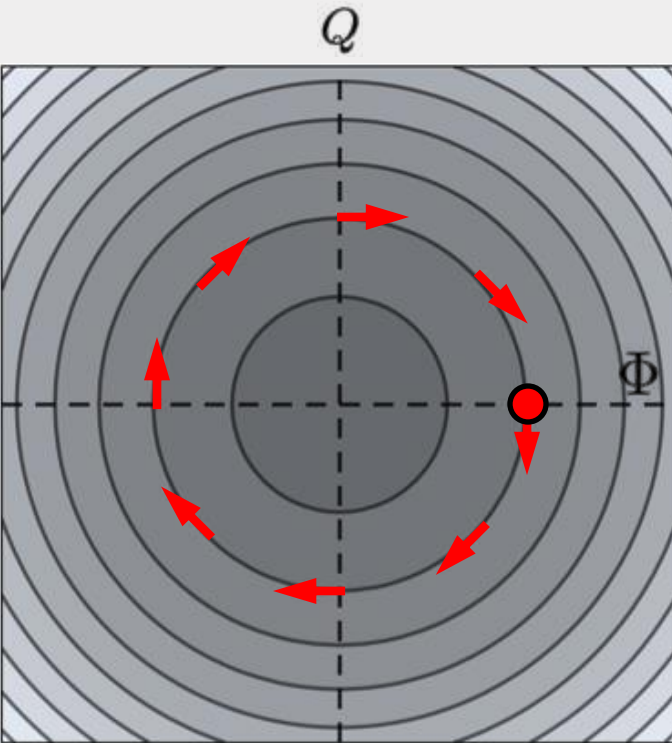


Phase-space (Husimi Q function)



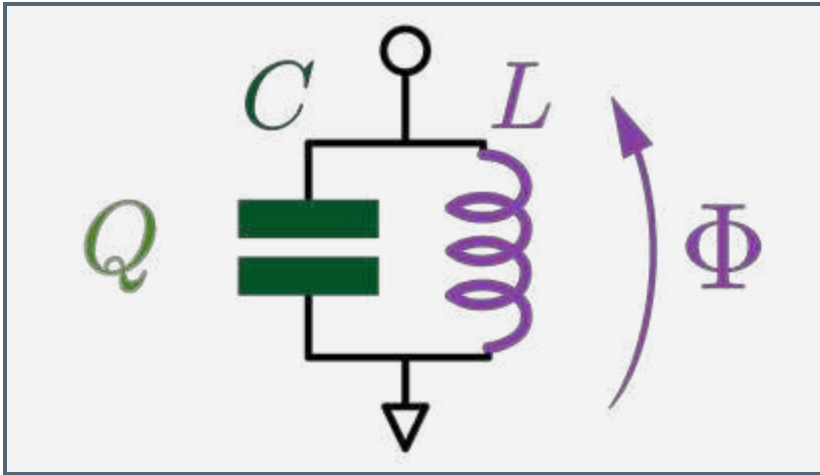
$|0\rangle$
 Q

$|1\rangle$
 Q

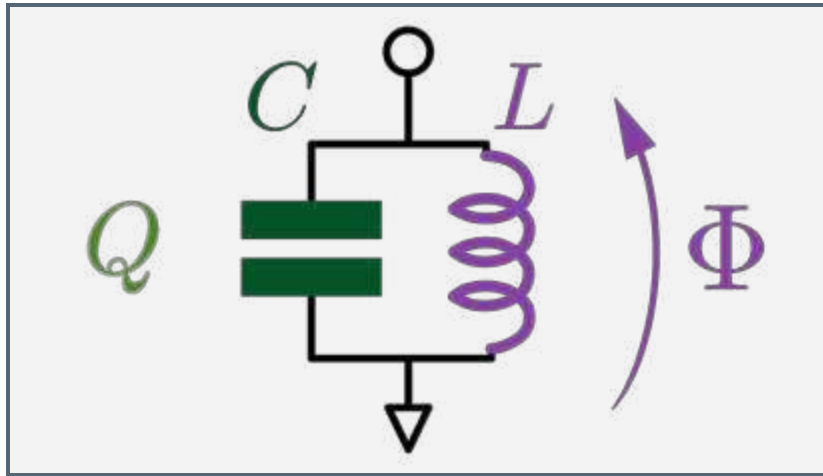


$$Q(\alpha) = \frac{1}{\pi} \langle \alpha | \hat{\rho} | \alpha \rangle$$

Linear harmonic oscillator summary



The LC quantum harmonic oscillator



Position: $\Phi \mapsto x$

Mass: $C \mapsto m$

Momentum: $Q \mapsto p$
Spring constant: $L^{-1} \mapsto k$

$$\hat{H}(\hat{\Phi}, \hat{Q}) = \frac{\hat{\Phi}^2}{2L} + \frac{\hat{Q}^2}{2C} = \hbar\omega_0 \left(\hat{a}^\dagger \hat{a} + \frac{1}{2} \right) \quad \omega_0^2 = \frac{1}{LC}$$

$$\hat{H} |n\rangle = \hbar\omega_0 \left(n + \frac{1}{2} \right) |n\rangle \quad Z_0 = \frac{L}{C}$$

$$\begin{aligned} \hat{\Phi} &= \Phi_{\text{ZPF}} (\hat{a}^\dagger + \hat{a}) & \Phi_{\text{ZPF}} &= \sqrt{\frac{\hbar}{2}} Z_0 = \Phi_0 \sqrt{\frac{z_0}{2\pi}}, \\ \hat{Q} &= iQ_{\text{ZPF}} (\hat{a}^\dagger - \hat{a}) & Q_{\text{ZPF}} &= \sqrt{\frac{\hbar}{2}} Z_0^{-1} = (2e) \sqrt{\frac{1}{2\pi z_0}}, \end{aligned}$$

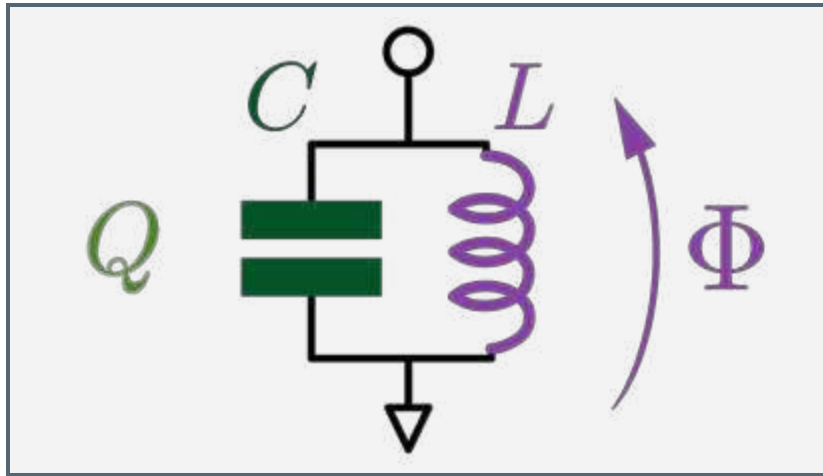
$$\mathcal{E}_{\text{ind}}(\Phi) = \frac{\Phi^2}{2L}$$

$$\begin{aligned} \mathcal{E}_{\text{cap}}(Q) &= \frac{Q^2}{2C} \\ &= \frac{1}{2} C \dot{\Phi}^2 \end{aligned}$$

Energy

Φ/ϕ_0

The LC quantum harmonic oscillator



Position: $\Phi \mapsto x$

Mass: $C \mapsto m$

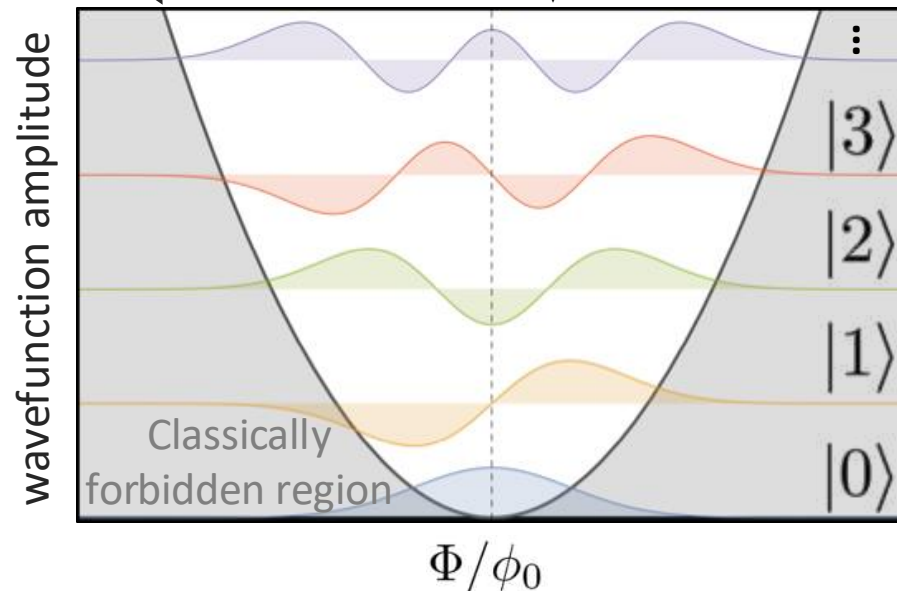
Momentum: $Q \mapsto p$
Spring constant: $L^{-1} \mapsto k$

$$\hat{H}(\hat{\Phi}, \hat{Q}) = \frac{\hat{\Phi}^2}{2L} + \frac{\hat{Q}^2}{2C} = \hbar\omega_0 \left(\hat{a}^\dagger \hat{a} + \frac{1}{2} \right) \quad \omega_0^2 = \frac{1}{LC}$$

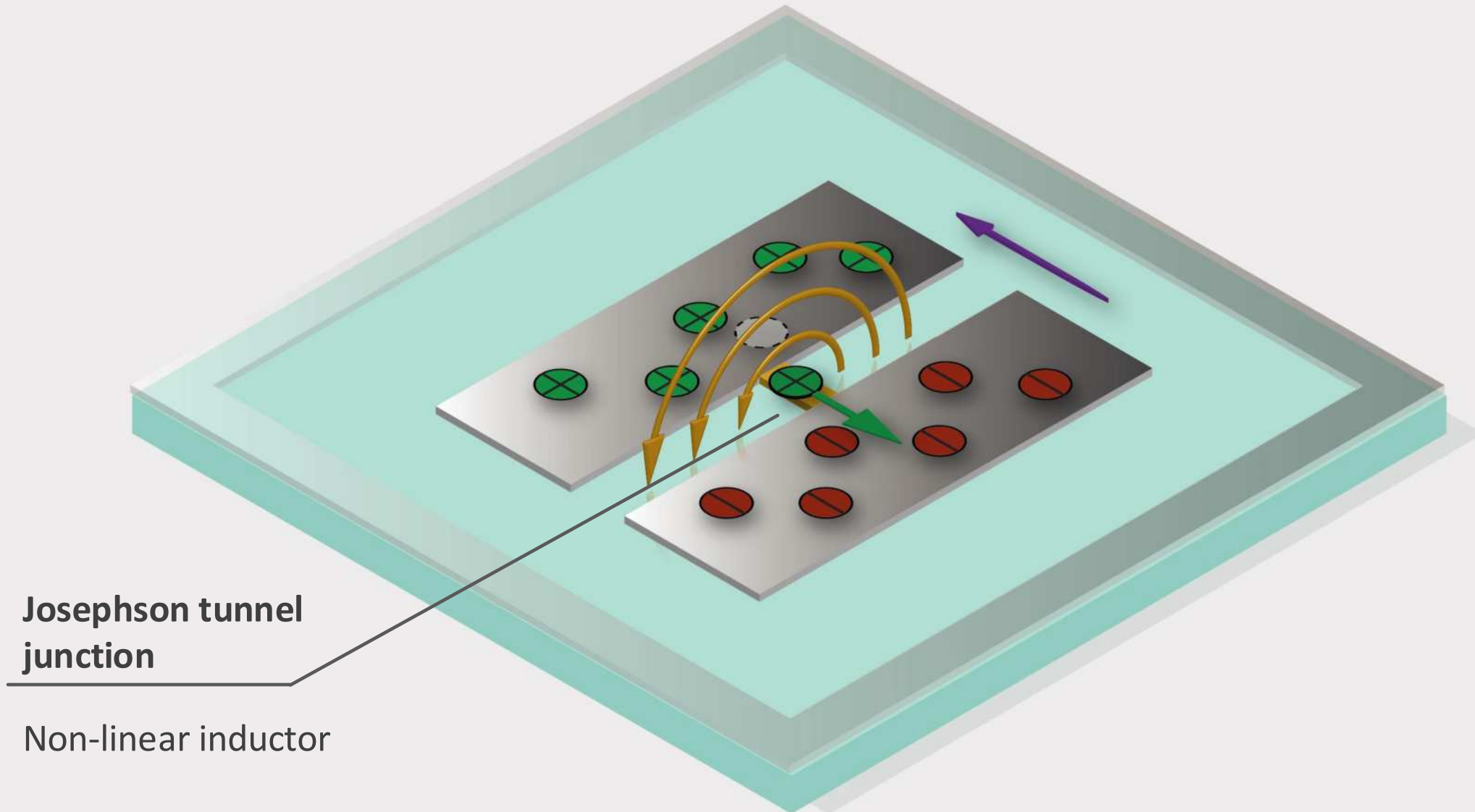
$$\hat{H}|n\rangle = \hbar\omega_0 \left(n + \frac{1}{2} \right) |n\rangle \quad \Phi_{\text{ZPF}} Q_{\text{ZPF}} = \frac{\hbar}{2} \quad Z_0 = \frac{L}{C}$$

$$\begin{aligned} \hat{\Phi} &= \Phi_{\text{ZPF}} (\hat{a}^\dagger + \hat{a}) & \Phi_{\text{ZPF}} &= \sqrt{\frac{\hbar}{2} Z_0} & \langle 0 | \hat{\Phi}^2 | 0 \rangle &= \Phi_{\text{ZPF}}^2 \\ \hat{Q} &= iQ_{\text{ZPF}} (\hat{a}^\dagger - \hat{a}) & Q_{\text{ZPF}} &= \sqrt{\frac{\hbar}{2} Z_0^{-1}} & \langle 0 | \hat{Q}^2 | 0 \rangle &= Q_{\text{ZPF}}^2 \end{aligned}$$

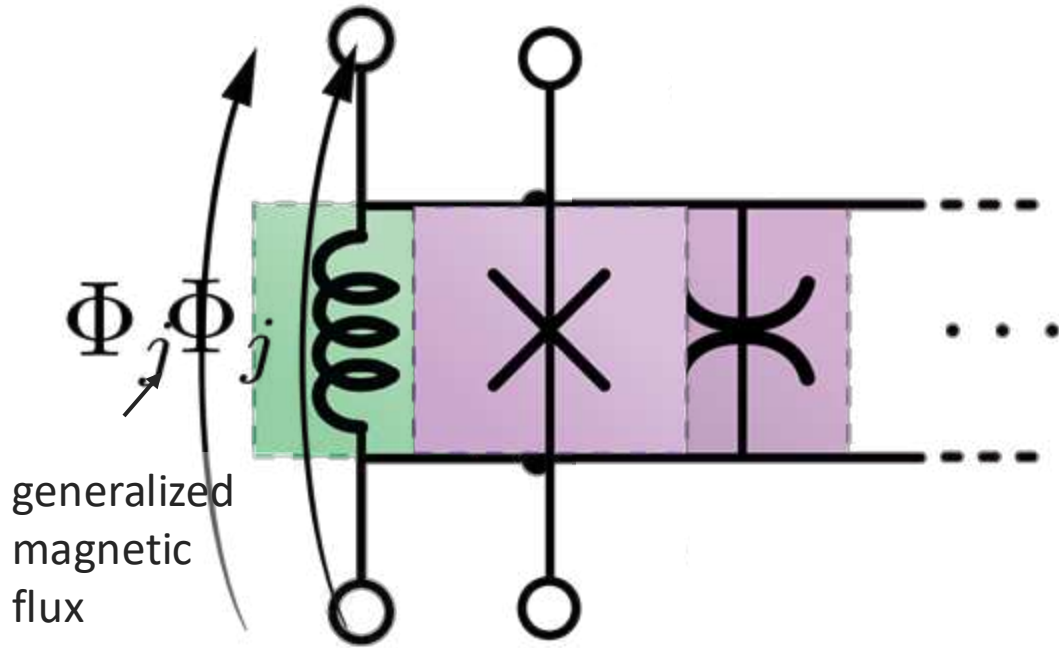
$$\mathcal{E}_{\text{ind}}(\Phi) = \frac{\Phi^2}{2L} \quad \psi_n(\Phi) \equiv \langle \Phi | n \rangle$$



The transmon as a non-linear oscillator



Josephson tunnel junction

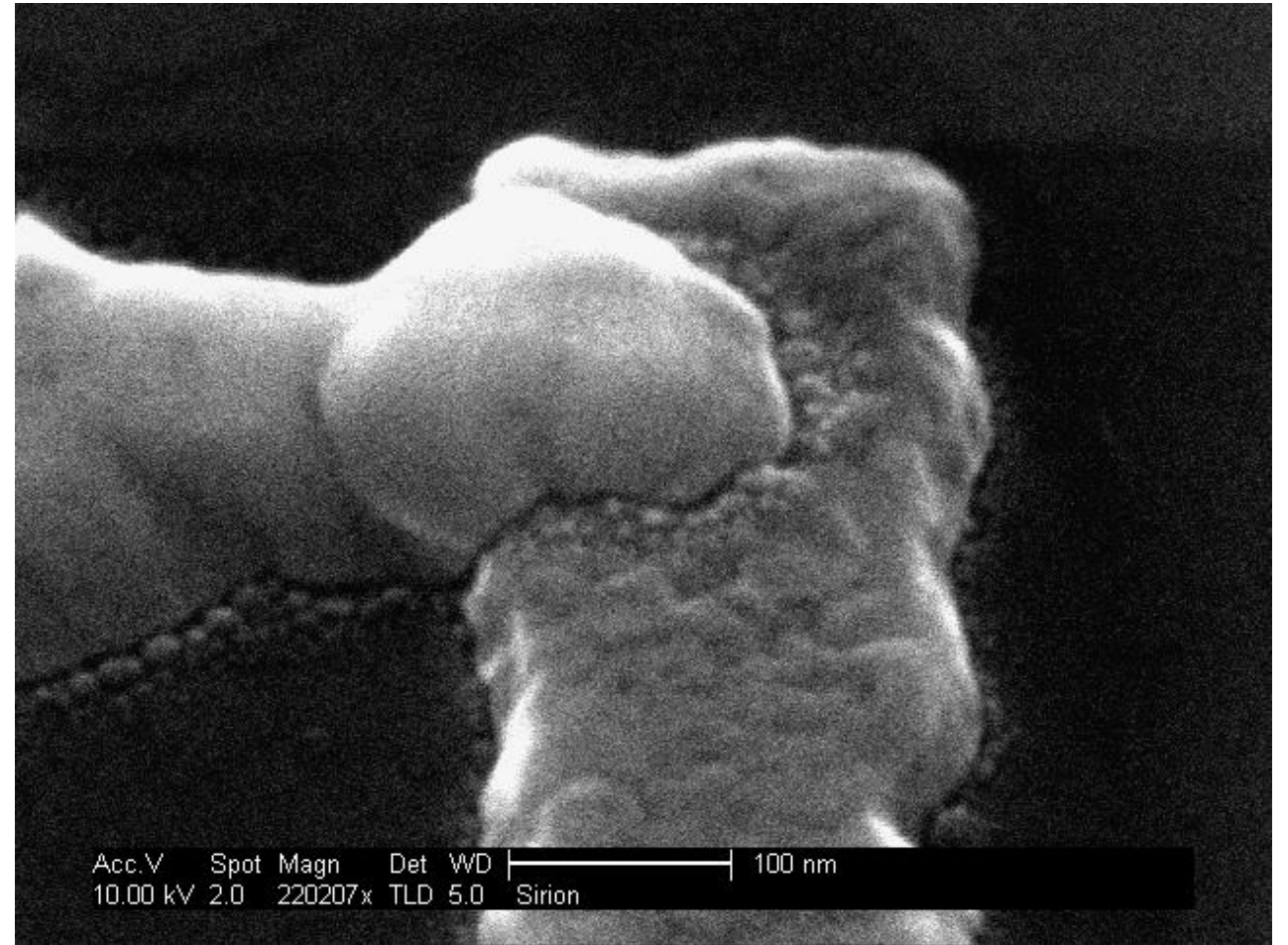


$$\mathcal{E}_j(\Phi_j) = E_j (1 - \cos(\Phi_j/\phi_0))$$

$$= \mathcal{E}_j^{\text{lin}}(\Phi_j) + \mathcal{E}_j^{\text{nl}}(\Phi_j)$$

$$\phi_0 \equiv \hbar/2e$$

$$= \frac{E_j}{2} \left(\frac{\Phi_j}{\phi_0} \right)^2 - \frac{E_j}{4!} \left(\frac{\Phi_j}{\phi_0} \right)^4 + \mathcal{O}(\Phi_j^6)$$

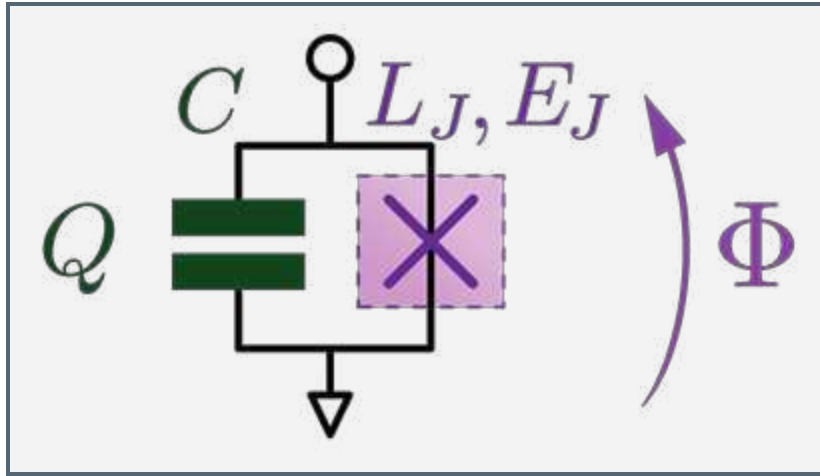


SEM image: L. Frunzio

Circuit image: Minev et al., EPR to appear (2020)

Zlatko Minev

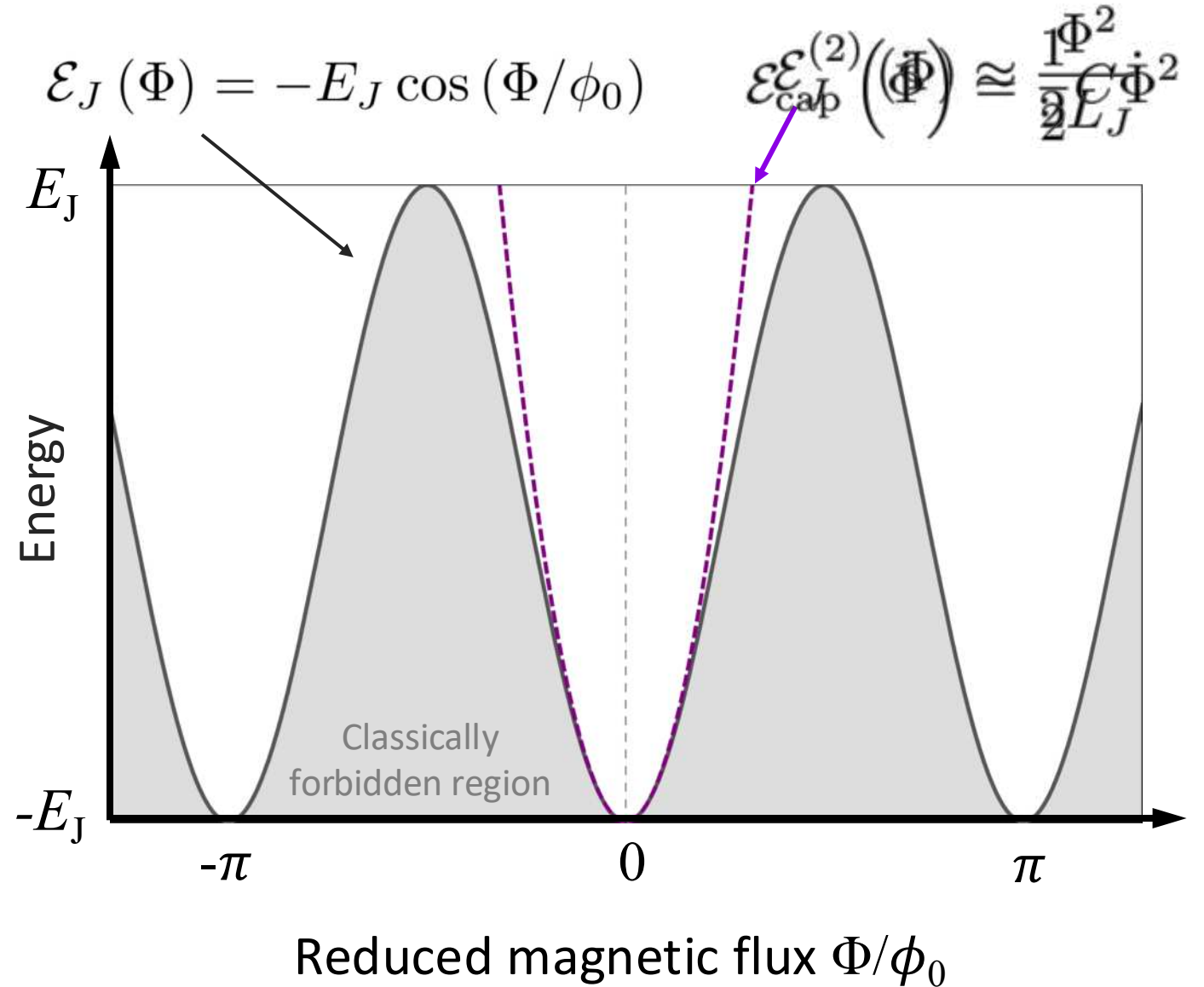
The Transmon qubit



$$E_J = \frac{\phi_0^2}{L_J}$$

$$\phi_0 = \frac{\hbar}{2e}$$

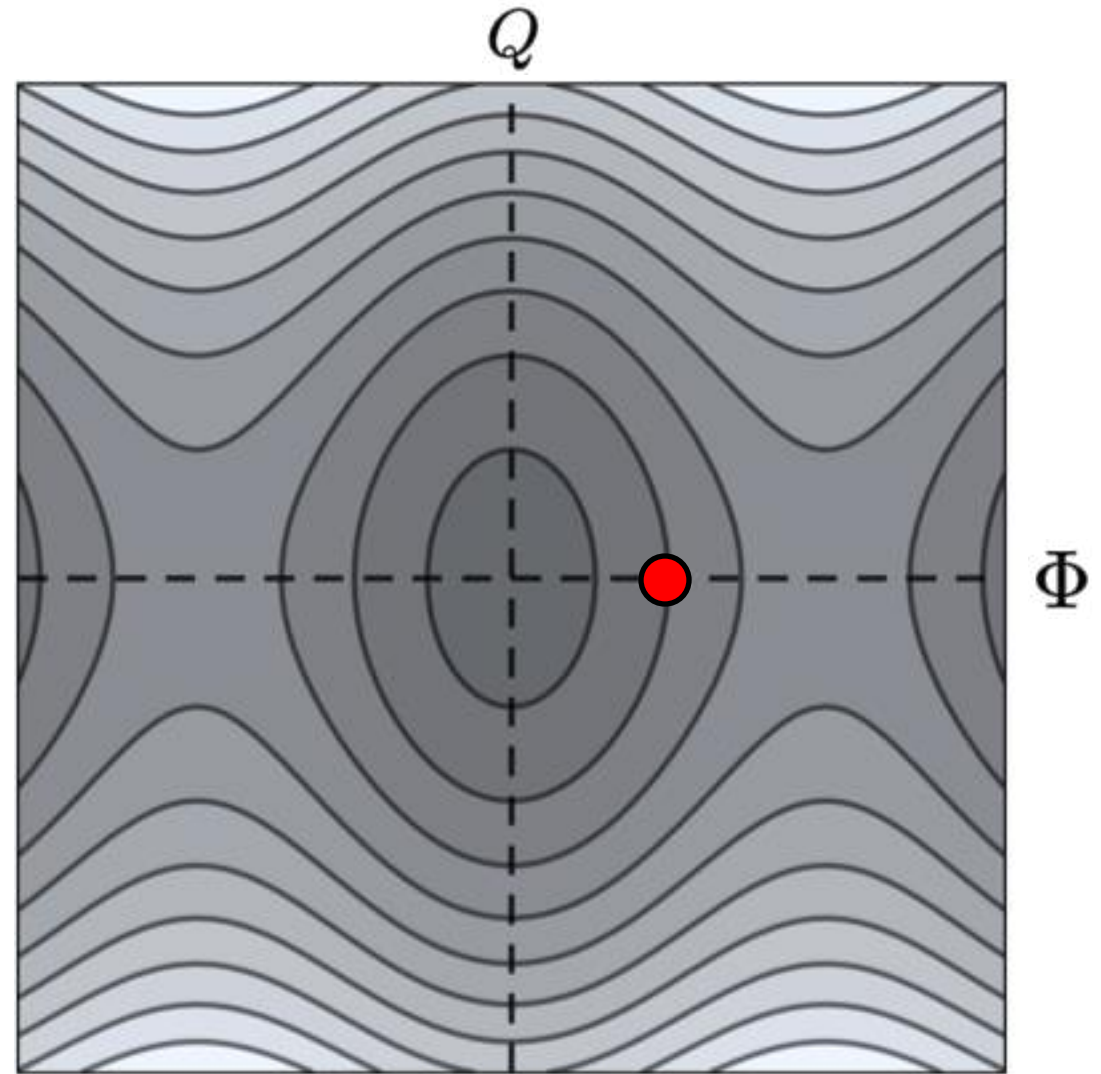
$$\approx 3.3 \times 10^{-16} \text{ Wb}$$



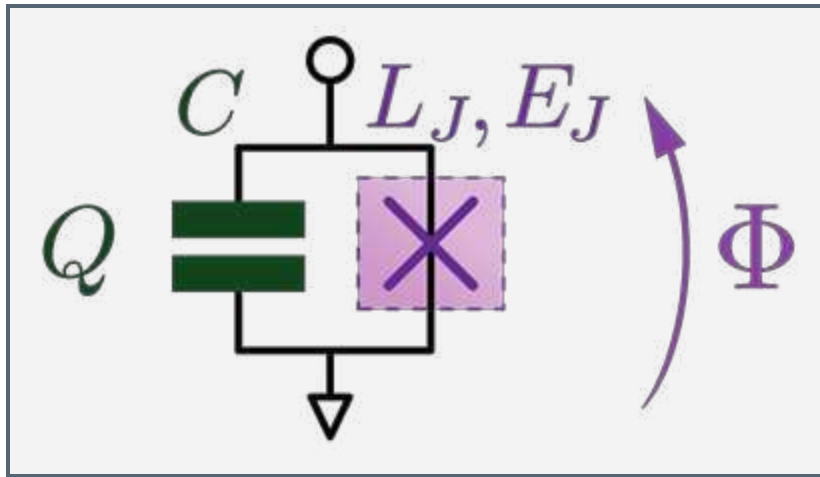
Semi-classical intuition: phase space picture

$$\mathcal{H}(\Phi, Q) = \frac{Q^2}{2C} - E_J \cos(\Phi/\phi_0)$$

$$\begin{aligned} E &= \hbar\omega_0 \left(n + \frac{1}{2} \right) \\ &= \frac{Q^2}{2C} - E_J \cos(\Phi/\phi_0) \end{aligned}$$



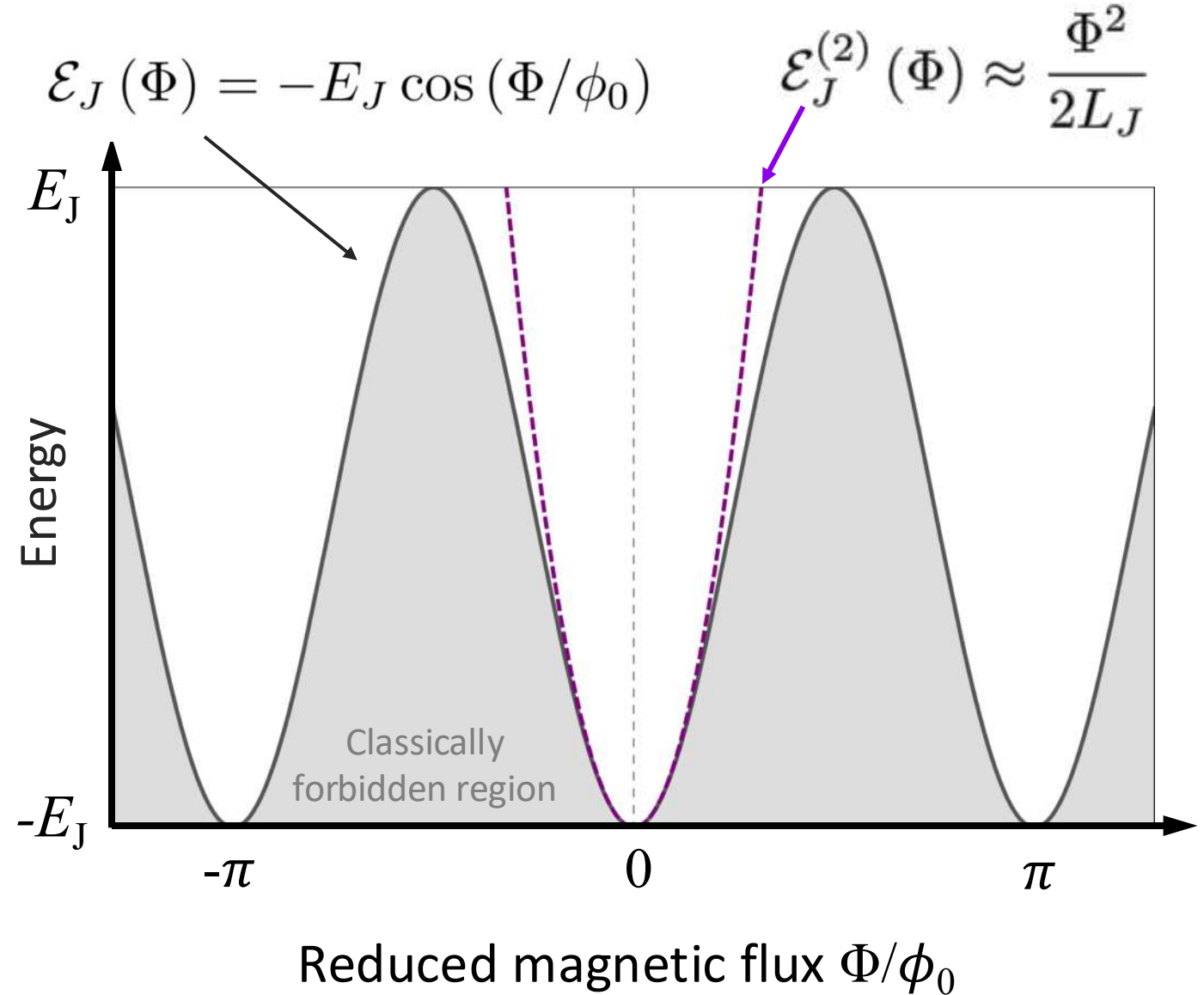
The Transmon qubit



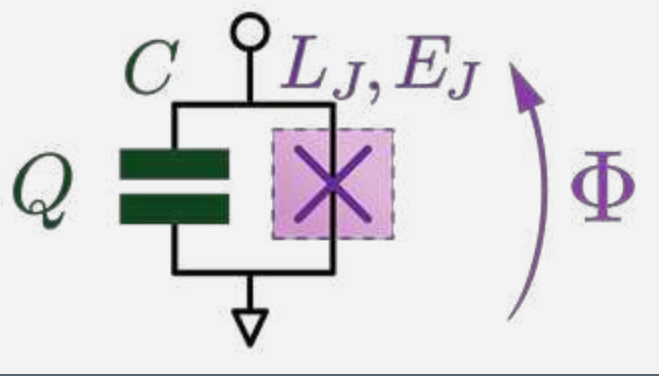
$$\hat{H} = \frac{\hat{Q}^2}{2C} - E_J \cos \left(\hat{\Phi} / \phi_0 \right)$$

$$\approx \frac{\hat{Q}^2}{2C} + \frac{\hat{\Phi}^2}{2L_J} - \frac{E_J}{4!} \left(\frac{\hat{\Phi}}{\phi_0} \right)^4$$

$$= \hbar \omega_0 \hat{a}^\dagger \hat{a} - \frac{E_J \phi_{\text{ZPF}}^4}{4!} (\hat{a} + \hat{a}^\dagger)^4$$



The Transmon qubit



$$= \hbar\omega_0 \hat{a}^\dagger \hat{a} - \frac{E_J \phi_{\text{ZPF}}^4}{4!} (3 + \cancel{6\hat{a}^2} + 12\hat{a}^\dagger \hat{a} + \cancel{6\hat{a}^{\dagger 2}} + \cancel{\hat{a}^4} + \cancel{4\hat{a}^\dagger \hat{a}^3} + \cancel{6\hat{a}^{\dagger 2} \hat{a}^2} + \cancel{4\hat{a}^{\dagger 3} \hat{a}} + \cancel{\hat{a}^{\dagger 4}})$$

RWA or 1st order PT

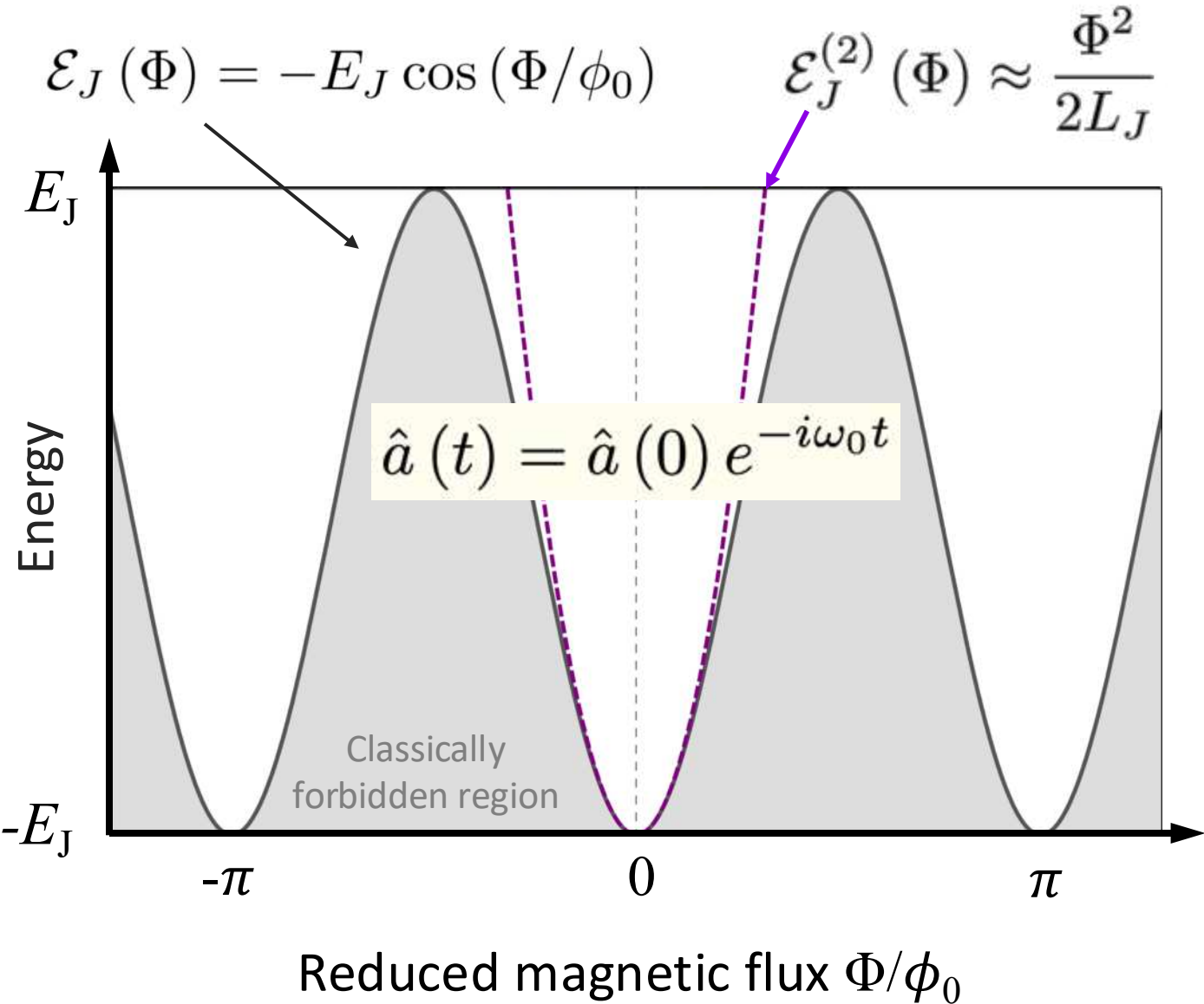
$$\hat{H}_4^{\text{RWA}} = \hbar\omega_0 \hat{a}^\dagger \hat{a} - \frac{E_J}{4!} \phi_a^4 (12\hat{a}^\dagger \hat{a} + 6\hat{a}^{\dagger 2} \hat{a}^2)$$

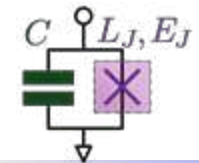
$$= \hbar(\omega_0 - \Delta_q) \hat{a}^\dagger \hat{a} - \frac{\hbar\alpha}{2} \hat{a}^{\dagger 2} \hat{a}^2$$

Due to nl & commutator Due to non-linearity (nl)

where $\hbar\Delta_q = \hbar\alpha \equiv \frac{1}{2} E_J \phi_{\text{ZPF}}^4$

“Lamb shift” due to ZPF Anharmonicity Zero-point fluctuations





The Kerr Hamiltonian of the transmon

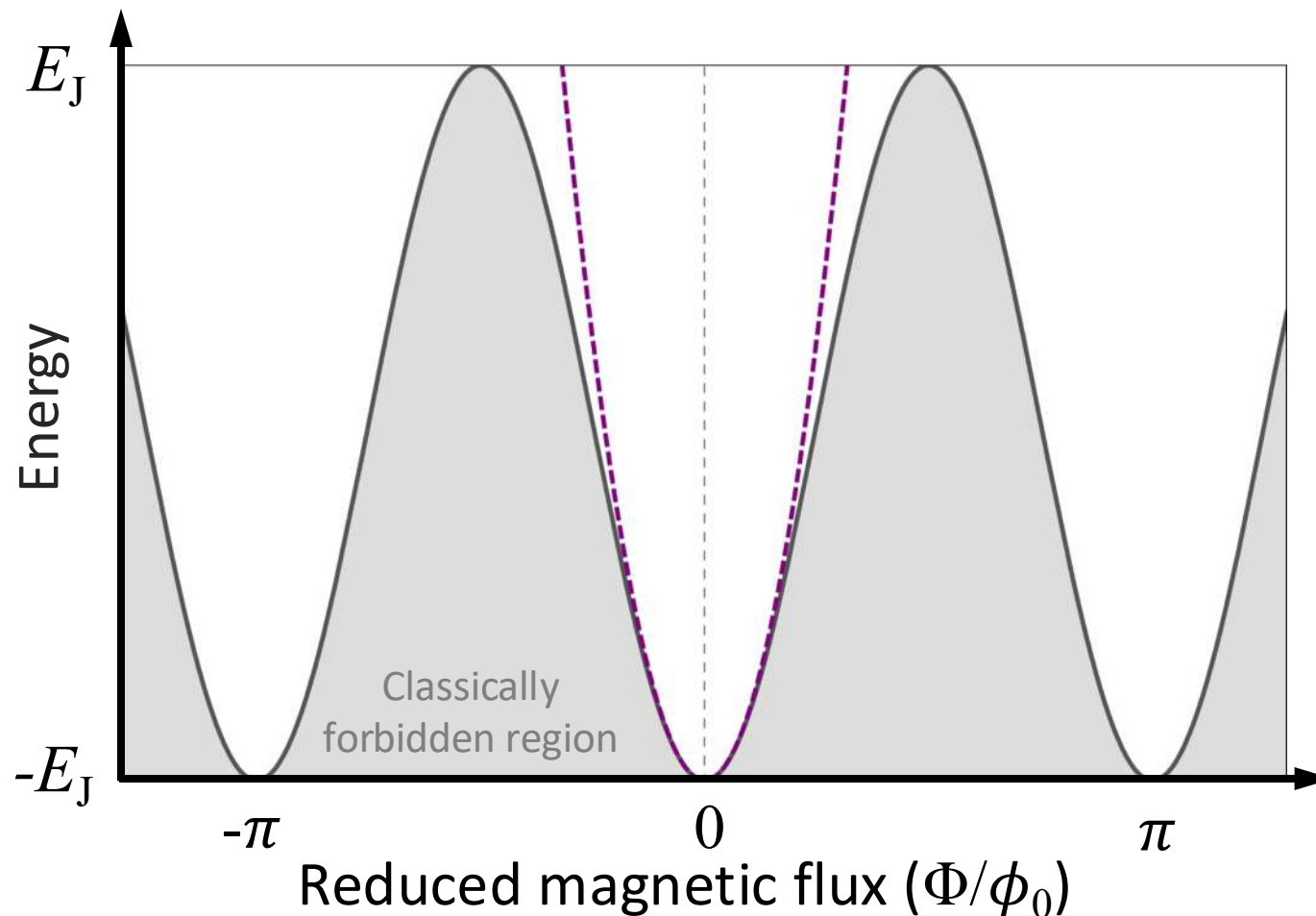
$$\hat{N} \equiv \hat{a}^\dagger \hat{a}$$

$$\hat{H}_4^{\text{RWA}} \approx \underbrace{\hbar\omega_q \hat{N}}_{\hat{H}_{\text{lin}}} - \underbrace{\frac{\hbar\alpha}{2} \hat{N} (\hat{N} - 1)}_{\hat{H}_{\text{nl}}}$$

$\hat{H}_{\text{lin}}$
 Solution known (SHO)

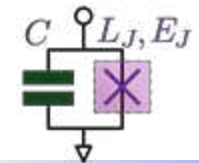
$\hat{H}_{\text{nl}}$
 New from nonlinearity

$$\hat{H}_4^{\text{RWA}} |n\rangle = \hbar\omega_q n \left(1 - \frac{\hbar\alpha_a}{2} (n - 1) \right) |n\rangle$$



1st order correction to energy: $E_n^{(1)} = \langle n^{(0)} | \hat{H}_{\text{nl}} | n^{(0)} \rangle$ and to eigenstates: $|n^{(1)}\rangle = \sum_{k \neq n} \frac{\langle k^{(0)} | \hat{H}_{\text{nl}} | n^{(0)} \rangle}{E_n^{(0)} - E_k^{(0)}} |k^{(0)}\rangle$.

To first order perturbation theory the eigenstates do not change! Only the energy changes. Dispersive.



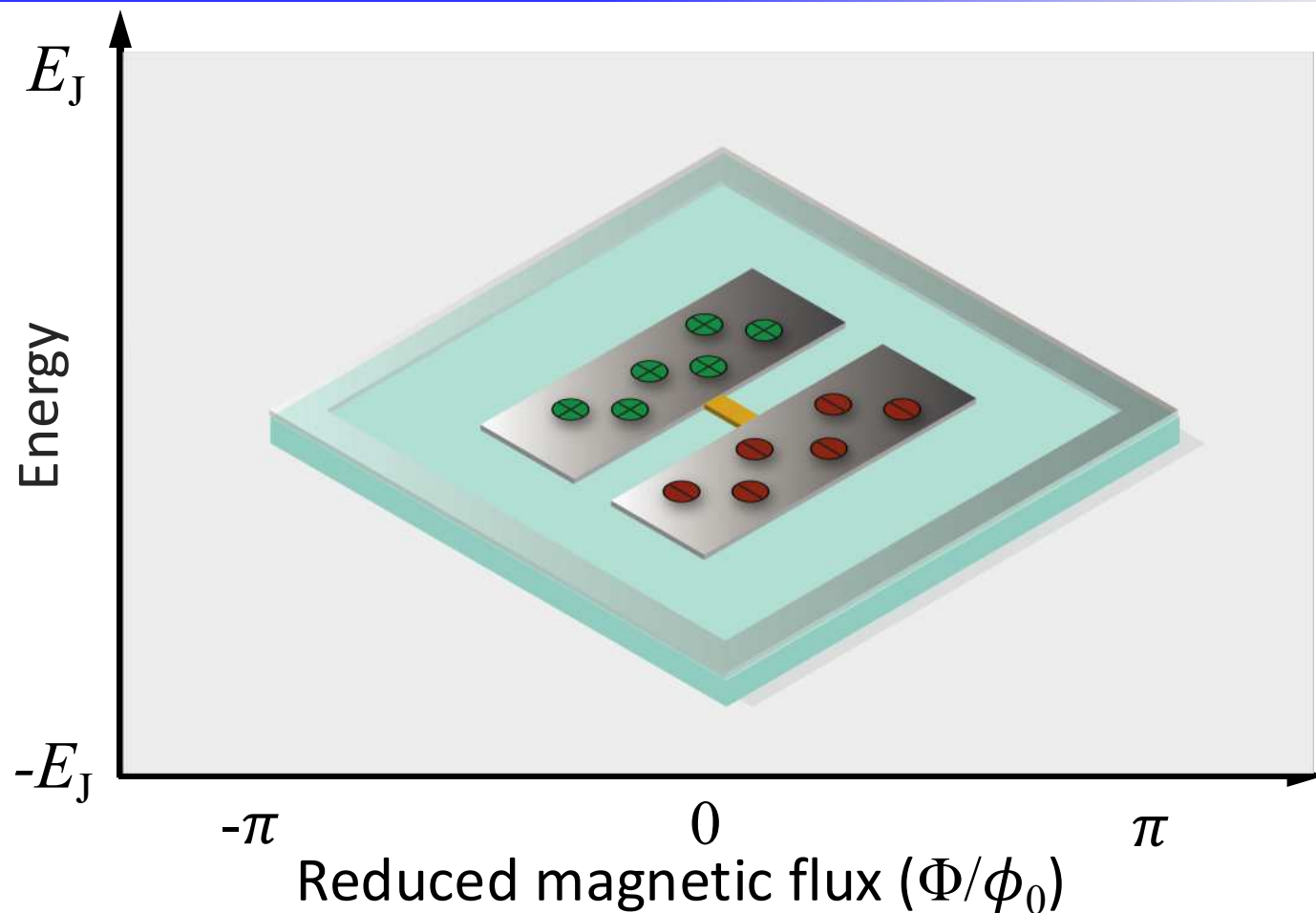
Exploring a real transmon qubit

$$\hat{N} \equiv \hat{a}^\dagger \hat{a}$$

$$\hat{H}_4^{\text{RWA}} \approx \underbrace{\hbar\omega_q \hat{N}}_{\hat{H}_{\text{lin}}} - \underbrace{\frac{\hbar\alpha}{2} \hat{N} (\hat{N} - 1)}_{\hat{H}_{\text{nl}}}$$

Solution known (SHO)

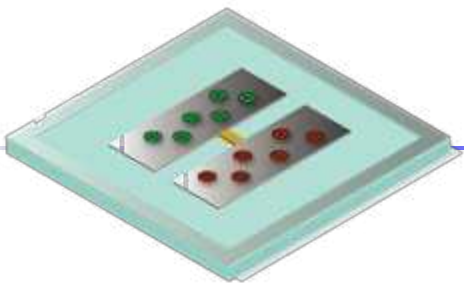
New from nonlinearity



Experimental parameters

$$L_J = 14 \text{ nH} \quad E_J = \frac{\phi_0^2}{L_J} = 12 \text{ GHz}$$

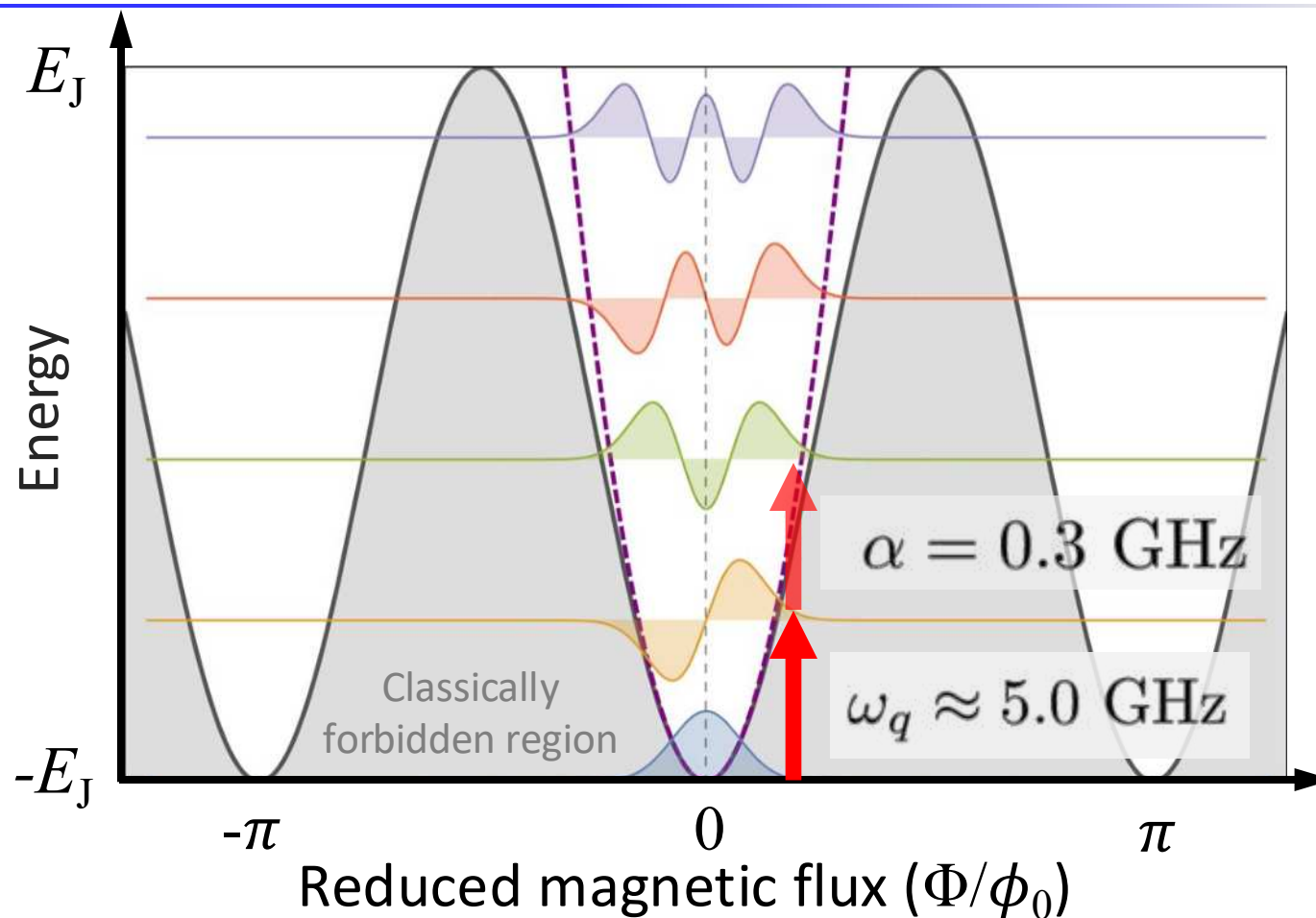
$$C_J = 65 \text{ fF} \quad E_C = \frac{e^2}{2C} = 0.3 \text{ GHz}$$



Exploring a real transmon qubit

$$\hat{H}_4^{\text{RWA}} \approx \hbar\omega_q \hat{N} - \frac{\hbar\alpha}{2} \hat{N} (\hat{N} - 1)$$

Dispersive, states didn't change to 1st order
“Lamb shift” due to ZPF



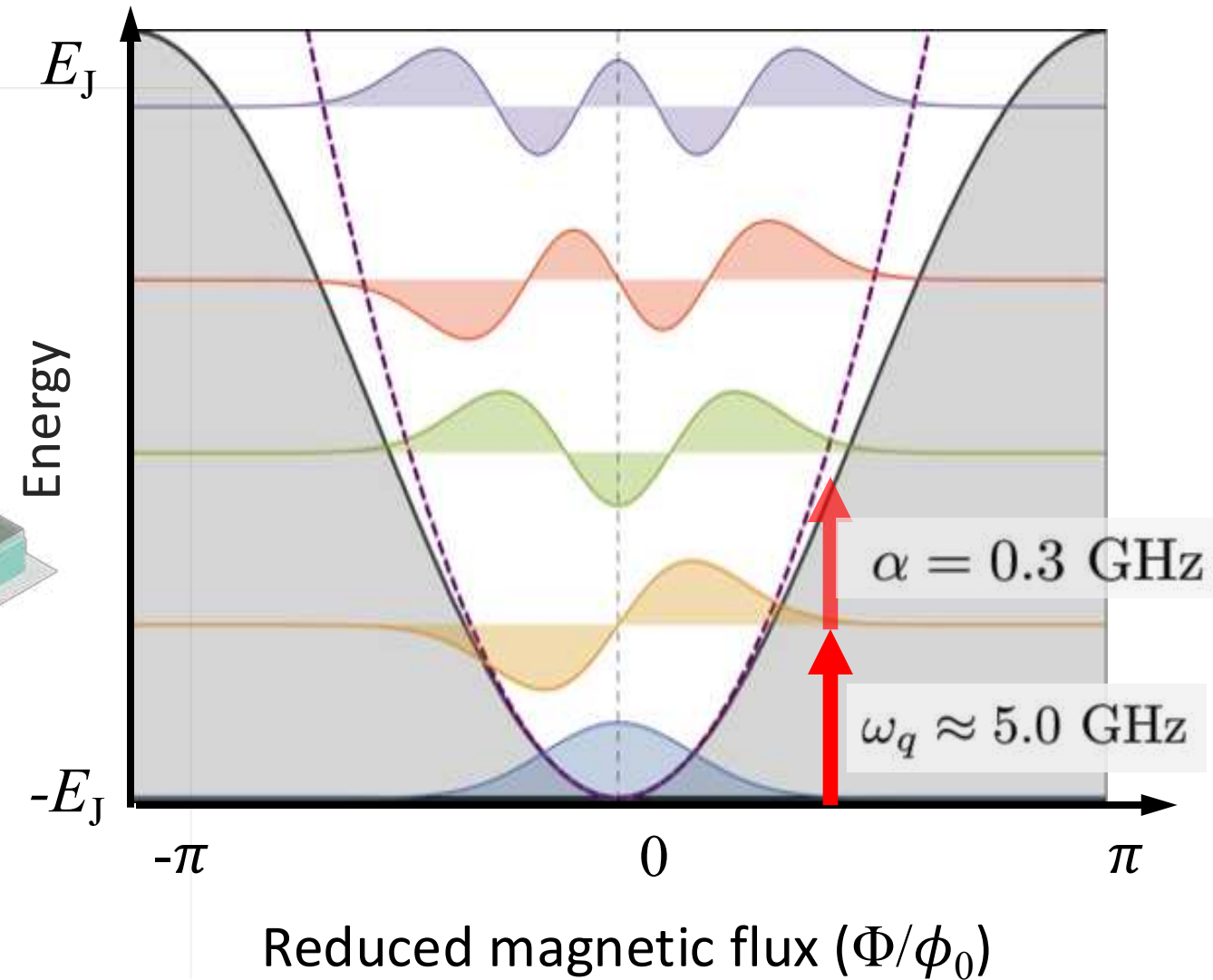
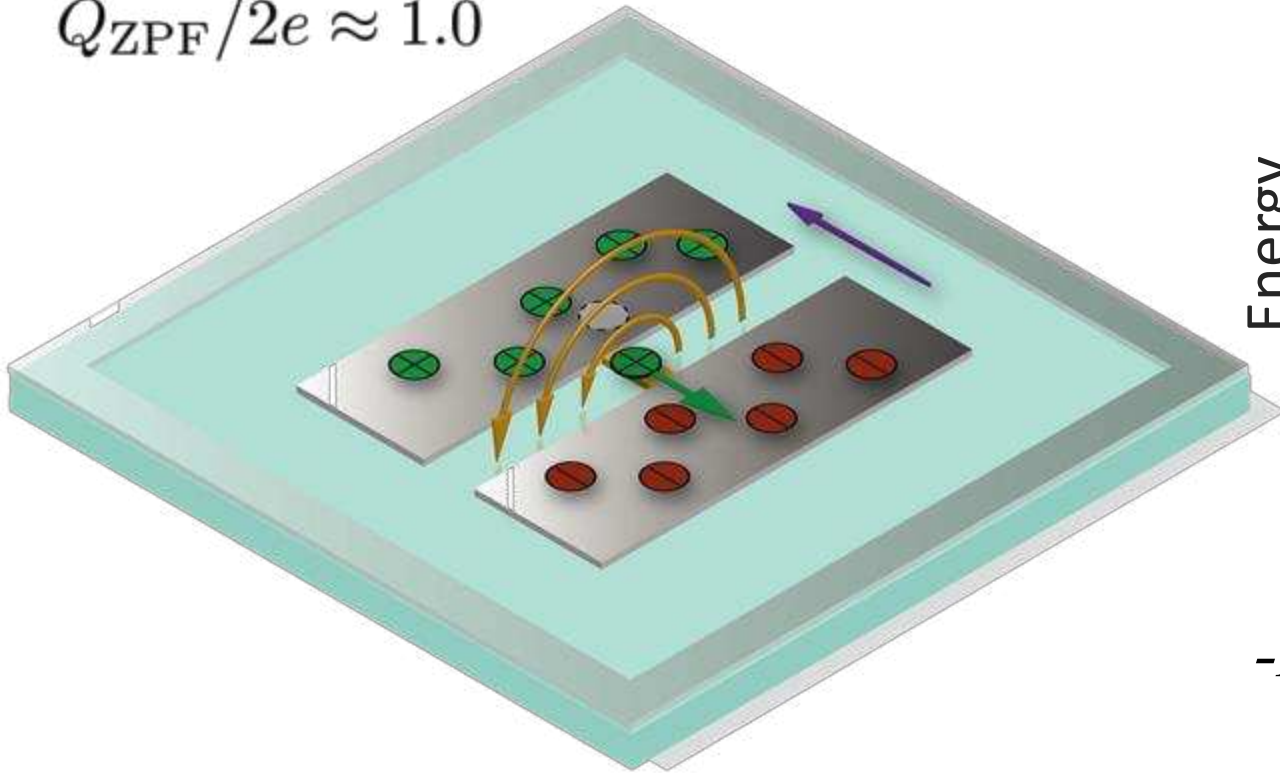
Parameters used in figure (of a measured qubit)

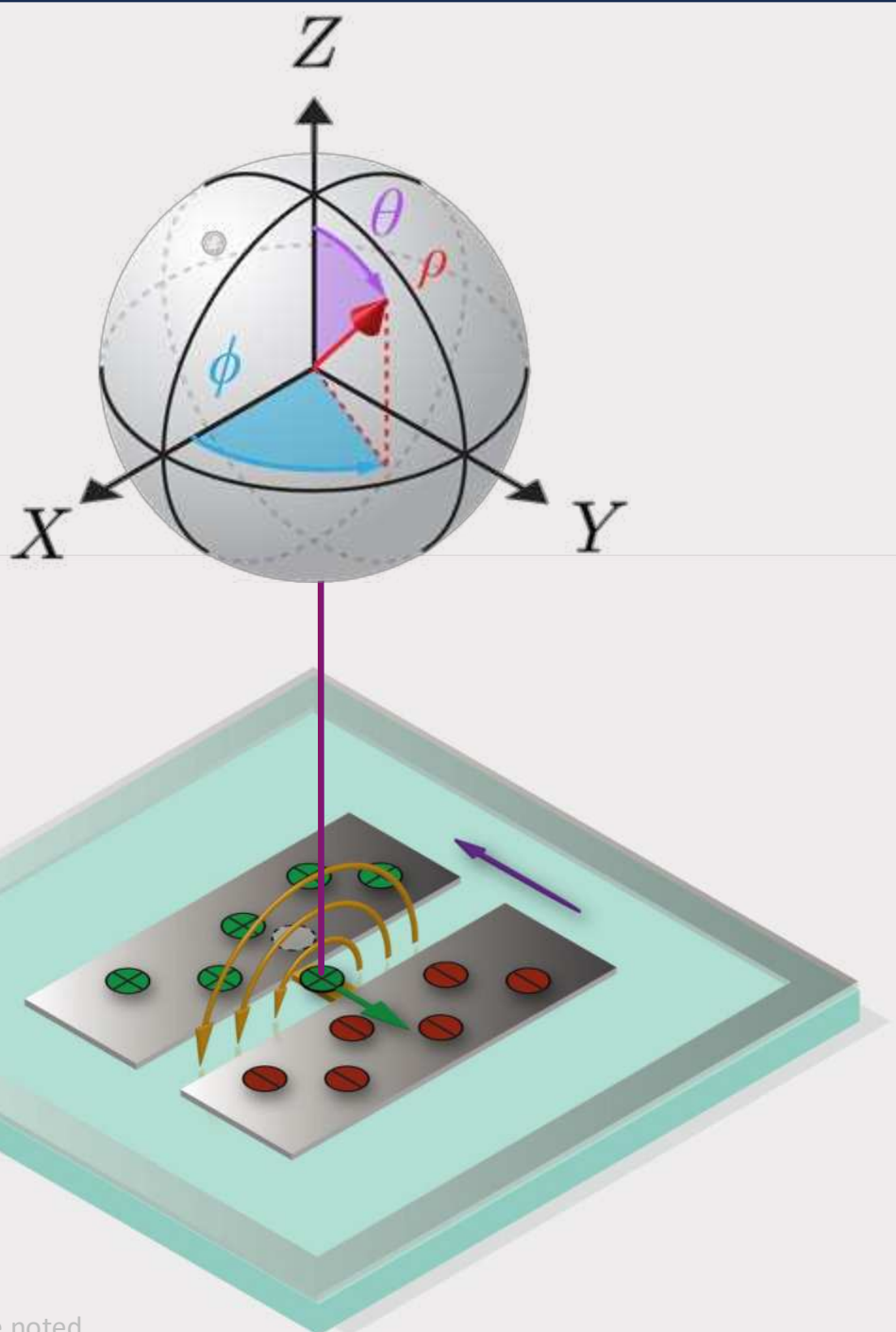
$$\begin{aligned} L_J &= 14 \text{ nH} & E_J &= \frac{\phi_0^2}{L_J} = 12 \text{ GHz} & \omega_0 &= \sqrt{\frac{1}{LC}} = 5.3 \text{ GHz} \\ C_J &= 65 \text{ fF} & E_C &= \frac{e^2}{2C} = 0.3 \text{ GHz} & Z &= \sqrt{\frac{L}{C}} \approx 450 \text{ } \Omega \end{aligned}$$

Quantum fluctuations of the transmon qubit

$$\Phi_{\text{ZPF}}/\phi_0 \approx 0.5$$

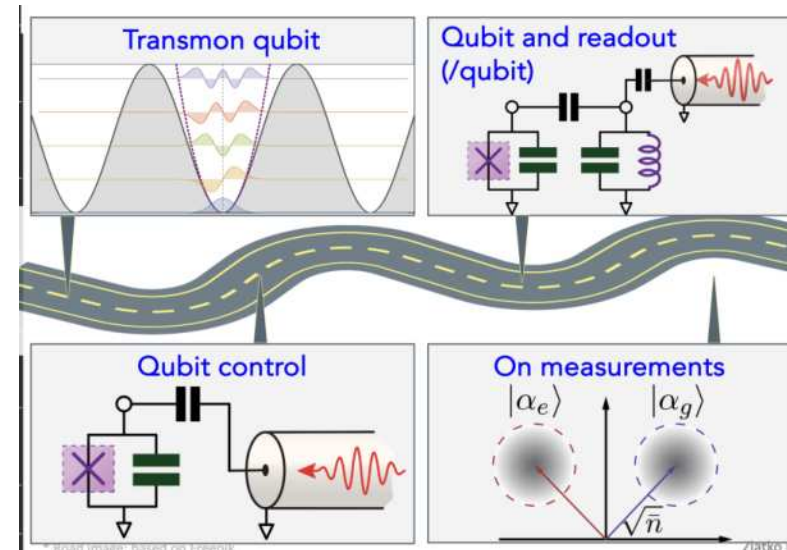
$$Q_{\text{ZPF}}/2e \approx 1.0$$





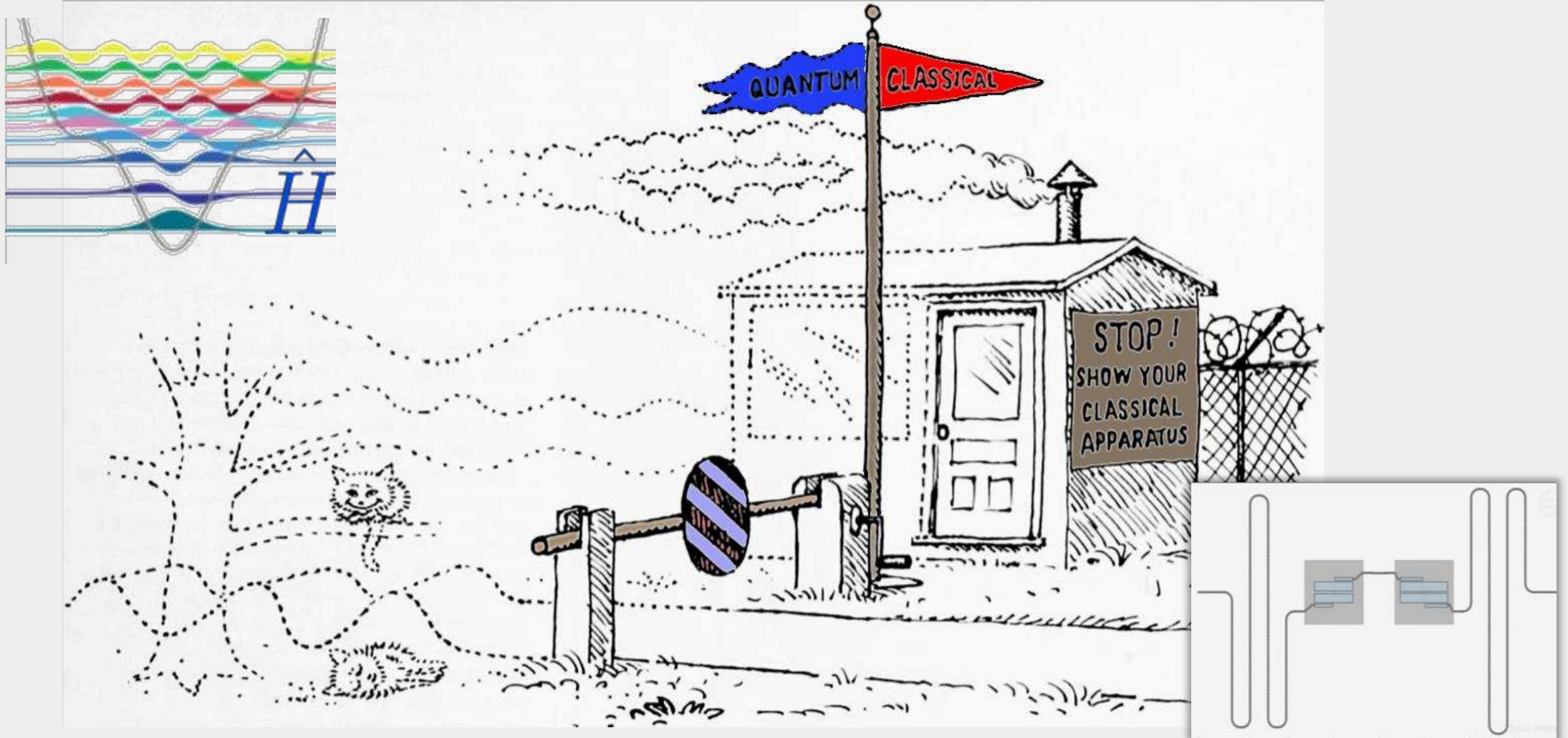
Advanced State of the Art Methods

Deeper dive on earlier material? See:

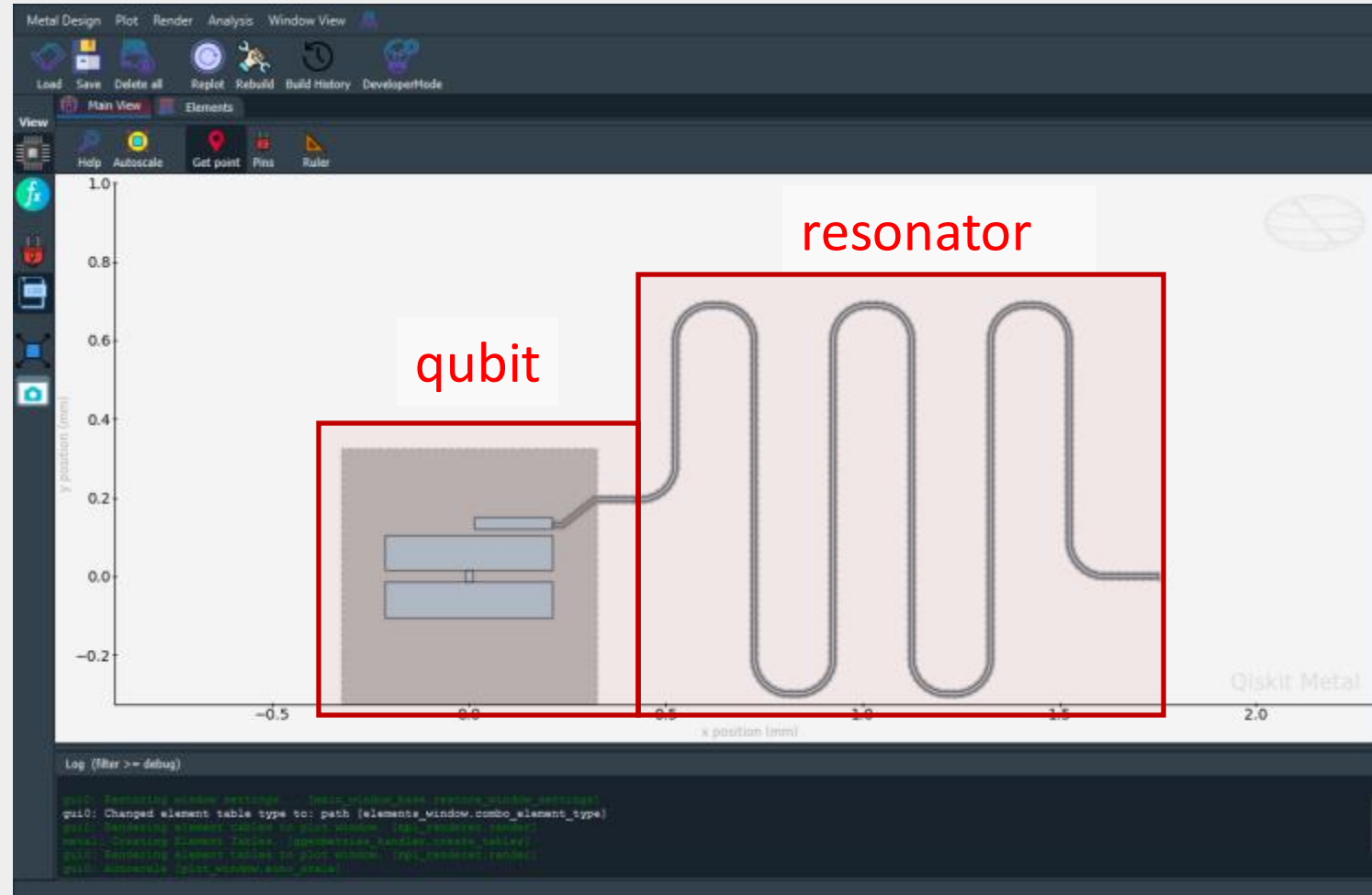


Slides!

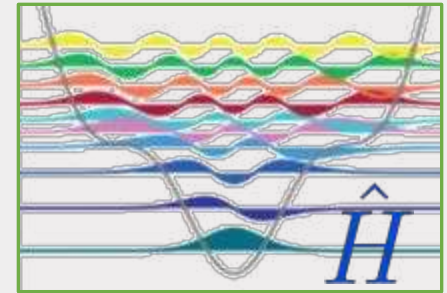
Quantum Analysis



Simple example

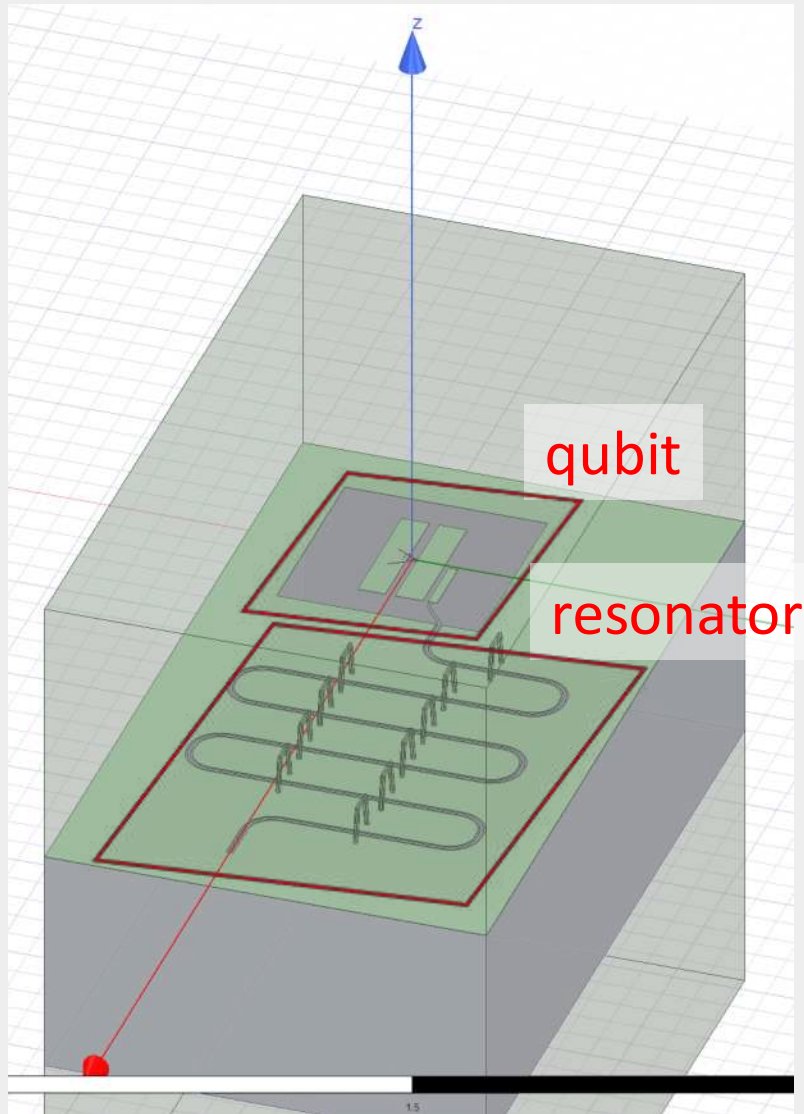


?

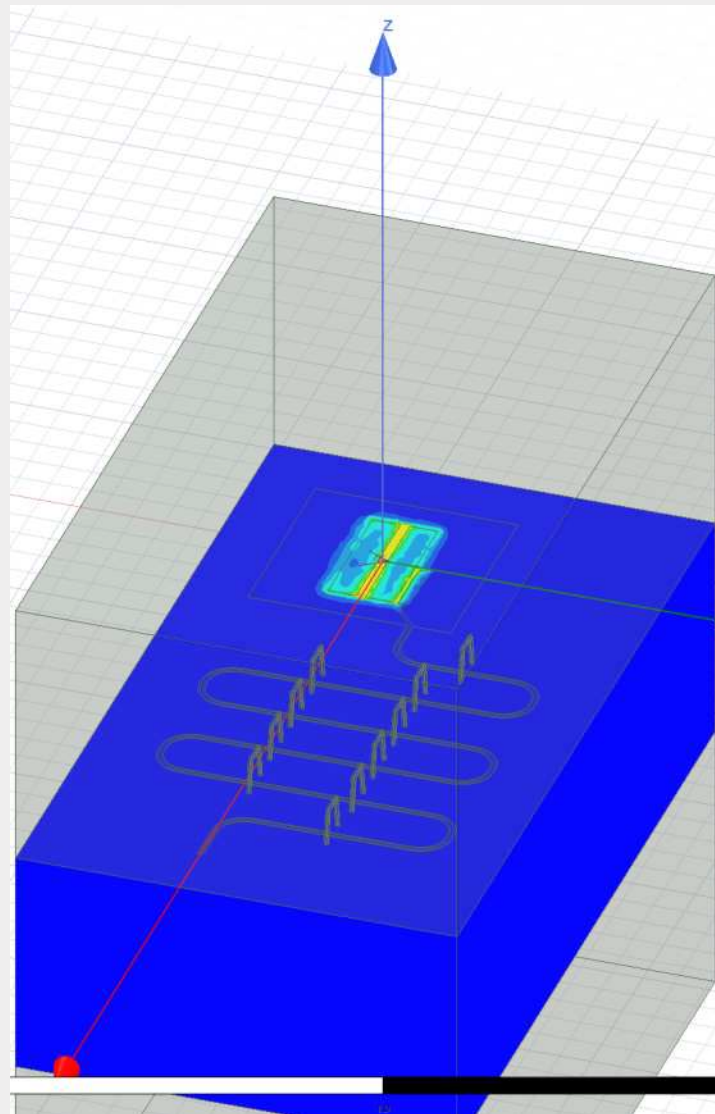


Modeling step 1: Classical E&M approximations

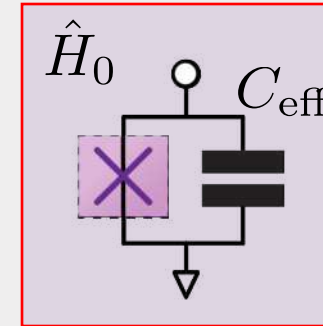
Physical (linear) model



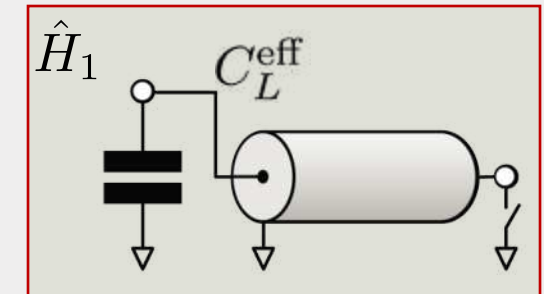
Eigenmodes



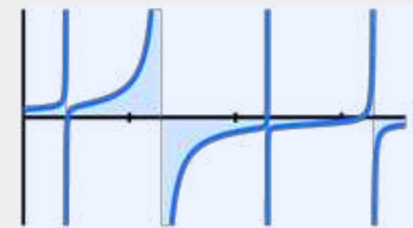
Quasi-lumped models



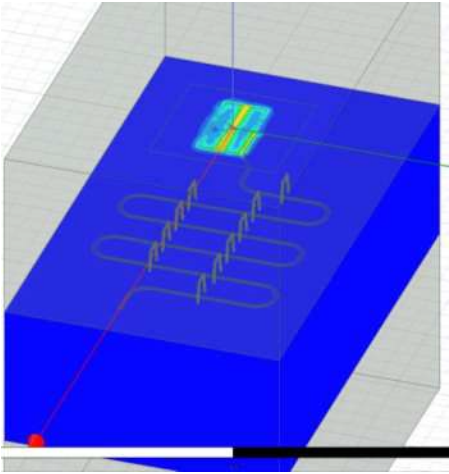
$$\begin{matrix} n_0 & n_1 & n_2 & n_3 & n_4 & n_5 \\ \begin{matrix} n_0 \\ n_1 \\ n_2 \\ n_3 \\ n_4 \\ n_5 \end{matrix} & \begin{pmatrix} C_{0\Sigma} & -C_{01} & -C_{02} & -C_{03} & -C_{04} & -C_{05} \\ & C_{1\Sigma} & -C_{12} & -C_{13} & -C_{14} & -C_{15} \\ & & C_{2\Sigma} & -C_{23} & -C_{24} & -C_{25} \\ & & & C_{3\Sigma} & -C_{34} & -C_{35} \\ & & & & C_{4\Sigma} & -C_{45} \\ & & & & & C_{5\Sigma} \end{pmatrix} \end{matrix}$$



Impedance



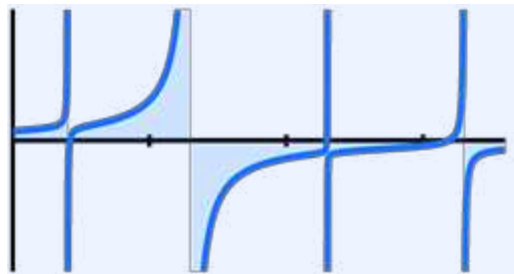
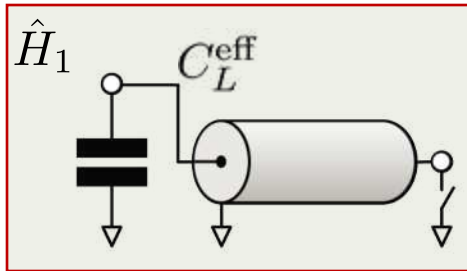
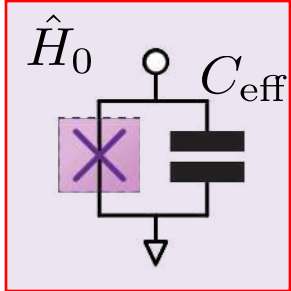
Modeling step 2: Quantize



$$\hat{H}_{\text{tot}}$$

$$\hat{H}_{\text{tot}} = \hat{H}_{\text{sys}} + \hat{H}_{\text{int}}$$

$$\hat{H}_{\text{tot}} = \hat{H}_{\text{lin}} + \hat{H}_{\text{nl}}$$



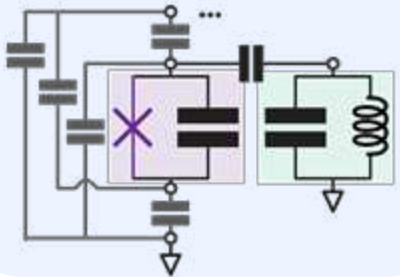
There's a description for every job!

Landscape of quantization methods

quasi-static

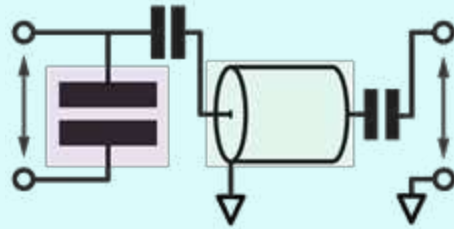
full-wave

lumped



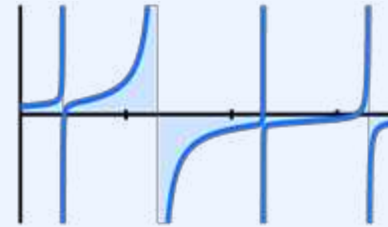
Yurke & Denker (1984), Devoret (1997),
Burkard *et al.* (2004), Koch *et al.*...

quasi-lumped



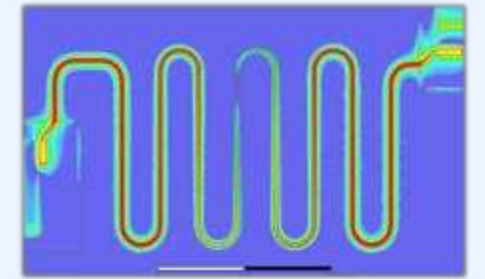
Malekakhlagh *et al.* (2017, 2019),
Gely *et al.* (2019), Parra-Rodriguez *et al.* (2019),
Minev *et al.* (2021), ...

impedance



Nigg *et al.* (2012), Bourassa *et al.* (2012),
Solgun *et al.* (2014, 2015, 2017) ...

energy



Minev (2018)
Minev *et al.* (2020)

more speed



more information, accuracy



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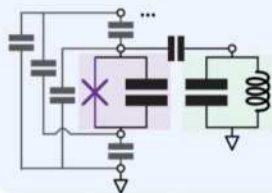


Quantization methods library

quasi-static

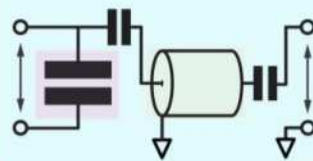
full-wave

lumped



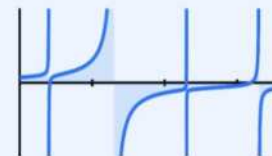
Yurke & Denker (1984), Devoret (1997), Burkard et al. (2004), Koch et al. ...

quasi-lumped



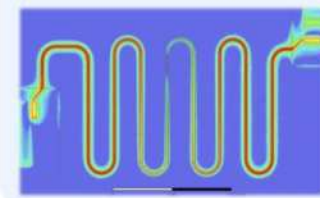
Malekakhlagh et al. (2017, 2019), Gely et al. (2019), Parra-Rodriguez et al. (2019), Minev et al. (2021), ...

impedance



Nigg et al. (2012), Bourassa et al. (2012), Solgun et al. (2014, 2015, 2017) ...

energy



Minev (2018)
Minev et al. (2020)

more speed



more information, accuracy



Zlatko Minev
IBM Quantum

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Quantization methods library: Tutorials

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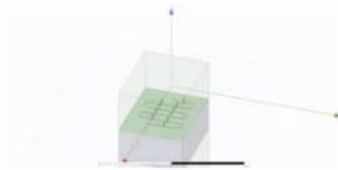
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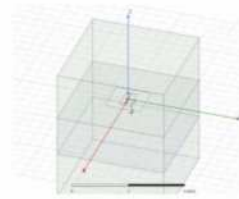
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Tutorials: Quantum analysis library



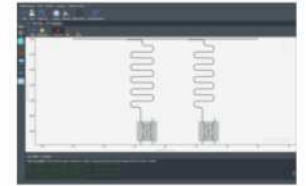
Eigenmode and EPR



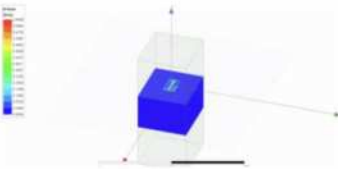
Impedance Analysis



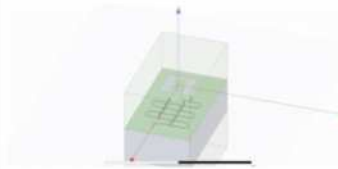
Capacitance Matrix and LOM Analysis



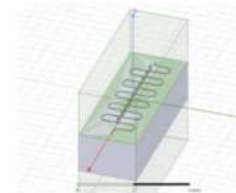
Sweep a option in QComponent, get Ansys HFSS driven-modal scatter matrix, impedance matrix, and admittance matrix.



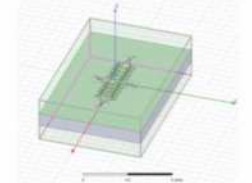
Analyzing and Tuning a Transmon Qubit



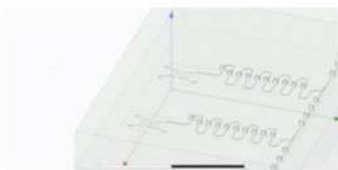
Analyzing a transmon and a resonator



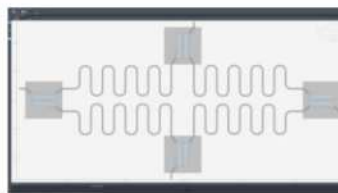
Analyzing a resonator



How to Render a Metal Design into Ansys



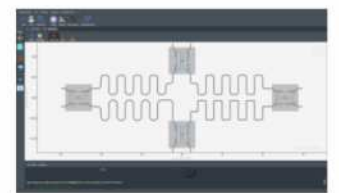
Driven Modal Simulation and S-Parameters



Sweep an option in QComponent, get capacitance matrix from Ansys Q3D and pyEPR



Sweep an option in QComponent, get Ansys HFSS eigenmode solution matrix.



How to Render a Metal Design into GDS

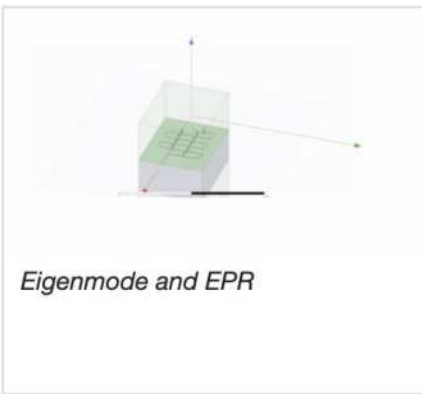
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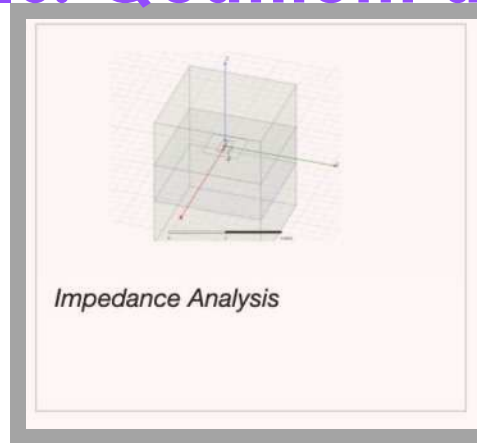
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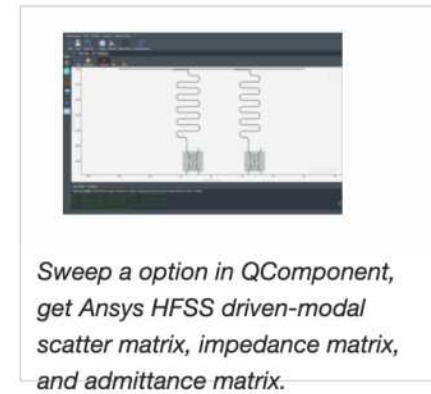
Eigenmode and EPR



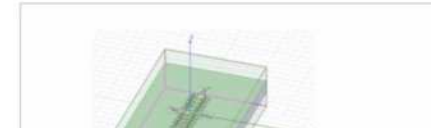
Impedance Analysis



Capacitance Matrix and LOM Analysis

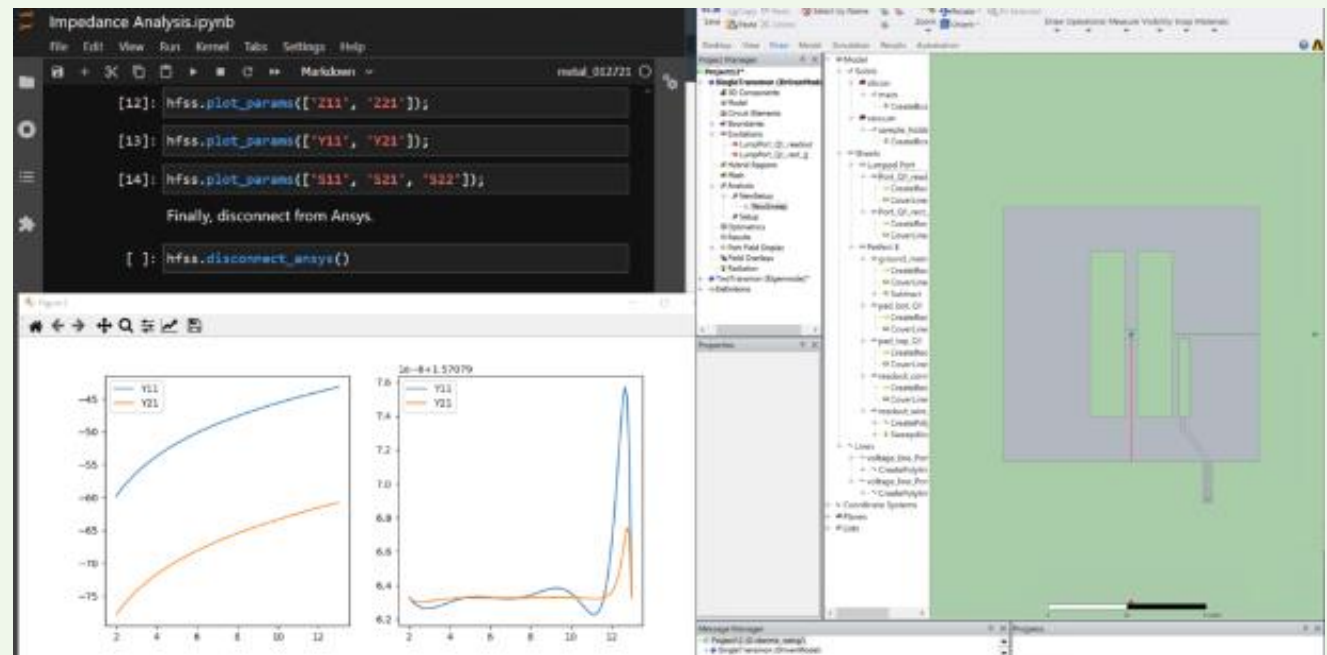


Sweep a option in QComponent, get Ansys HFSS driven-modal scatter matrix, impedance matrix, and admittance matrix.



S, Z, Y Impedance Scattering

arXiv:1204.0587 ...



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Eigenmode and EPR

Impedance Analysis

Capacitance Matrix and LOM
Analysis

Sweep a option in QComponent,
get Ansys HFSS driven-modal
scatter matrix, impedance matrix,
and admittance matrix.

Eigenmode

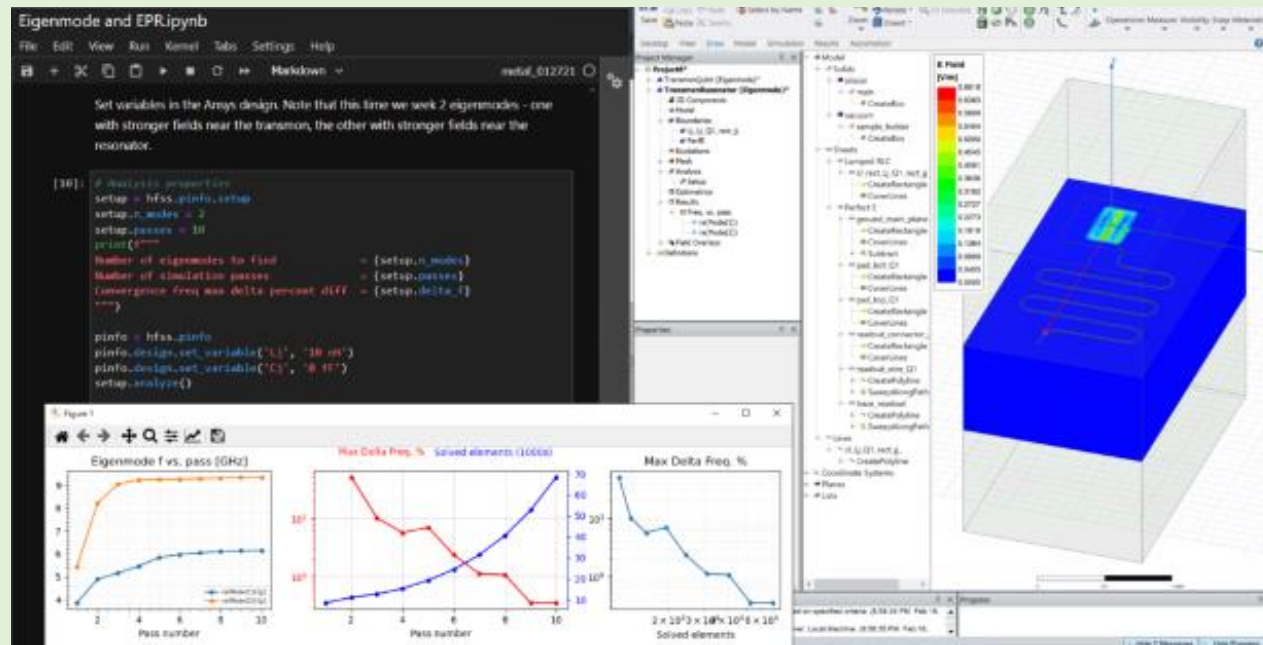
Energy-participation quantization of Josephson circuits

[Zlatko K. Minev](#) , [Zaki Leghtas](#), [Shantanu O. Mundhada](#), [Lysander Christakis](#), [Ioan M. Pop](#)
& [Michel H. Devoret](#)

[npj Quantum Information](#) **7**, Article number: 131 (2021) | [Cite this article](#)

[Minev arXiv: 1902.10355](#)

[Minev arXiv: 2010.00620](#)



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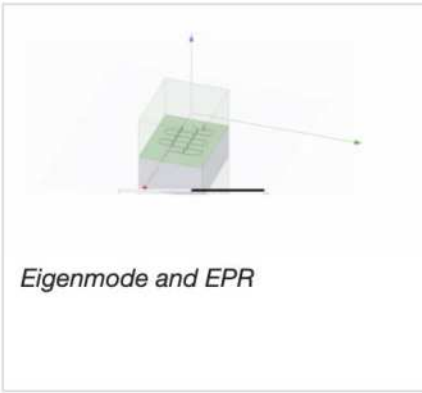
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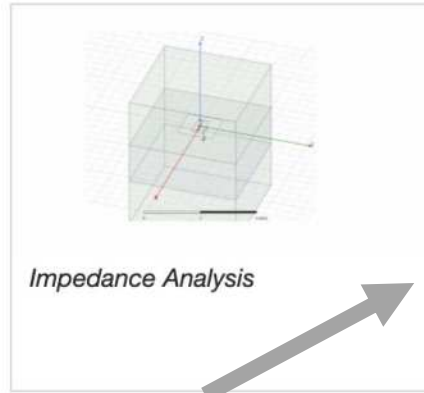
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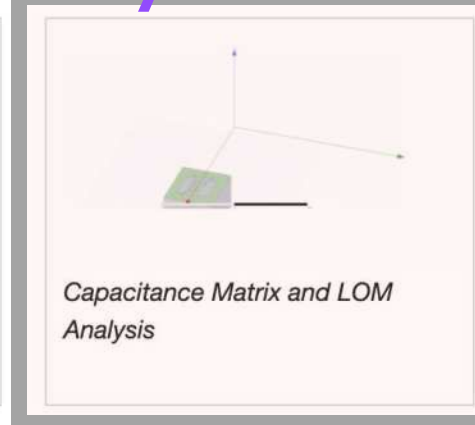
Tutorials: Quantum analysis library



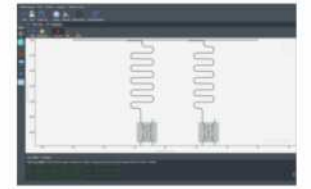
Eigenmode and EPR



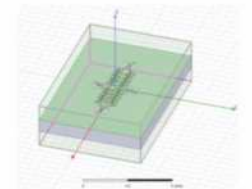
Impedance Analysis



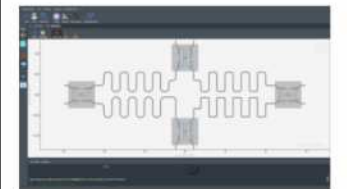
Capacitance Matrix and LOM Analysis



Sweep a option in QComponent, get Ansys HFSS driven-modal scatter matrix, impedance matrix, and admittance matrix.



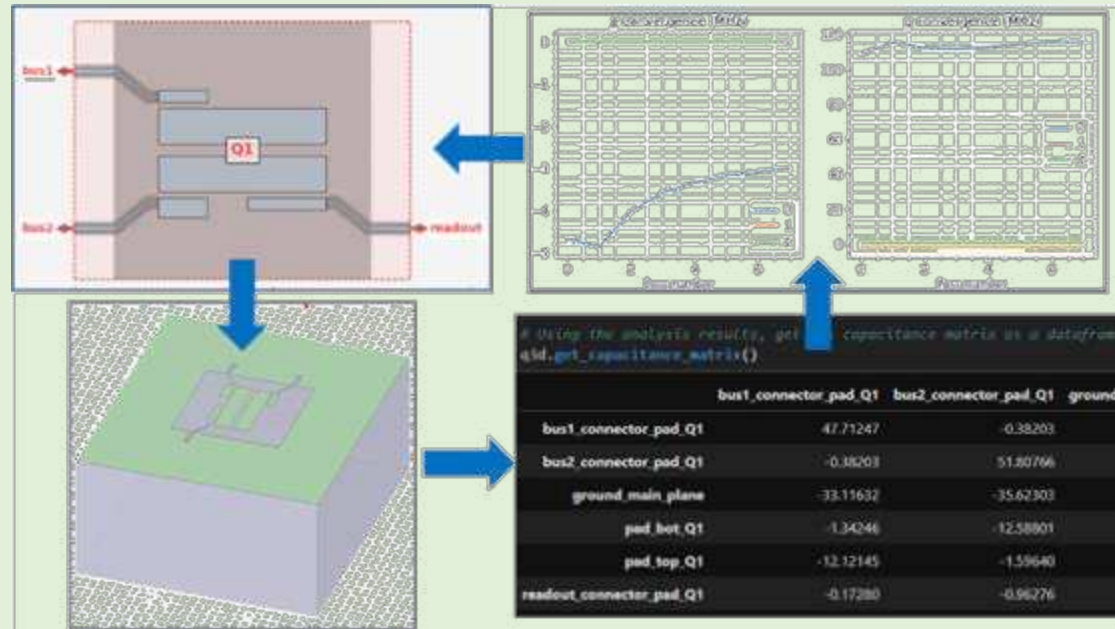
How to Render a Metal Design into Ansys



How to Render a Metal Design into GDS

Capacitive

arXiv:2103.10344 ...



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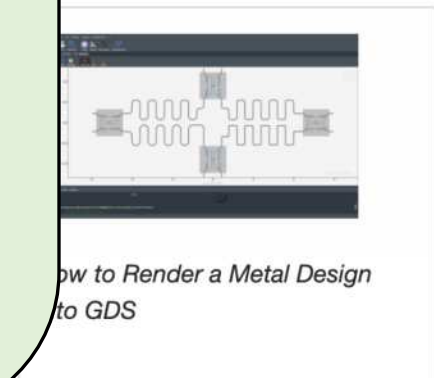
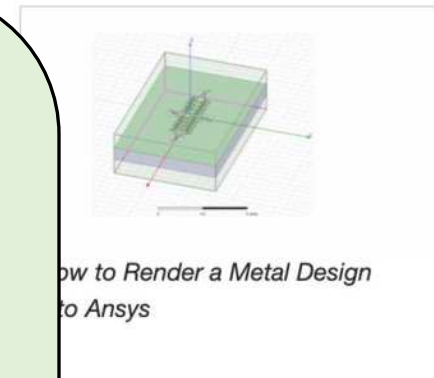
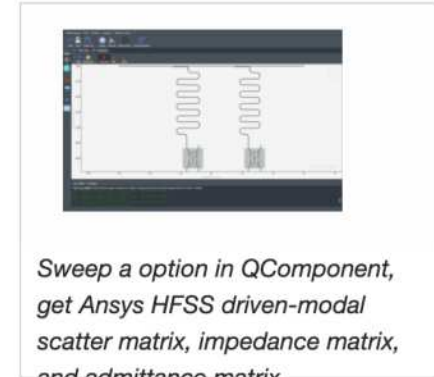
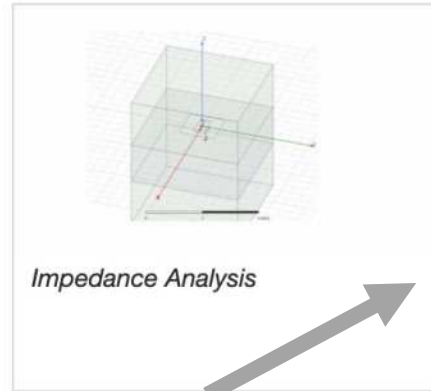
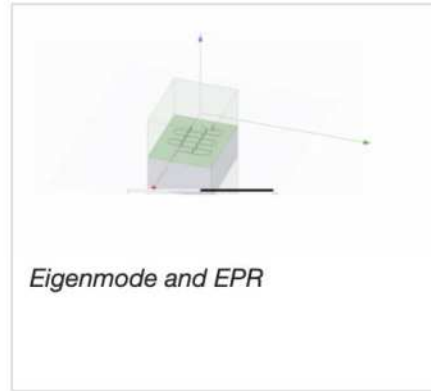
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Cornell University

arXiv.org > quant-ph > arXiv:2103.10344

Quantum Physics

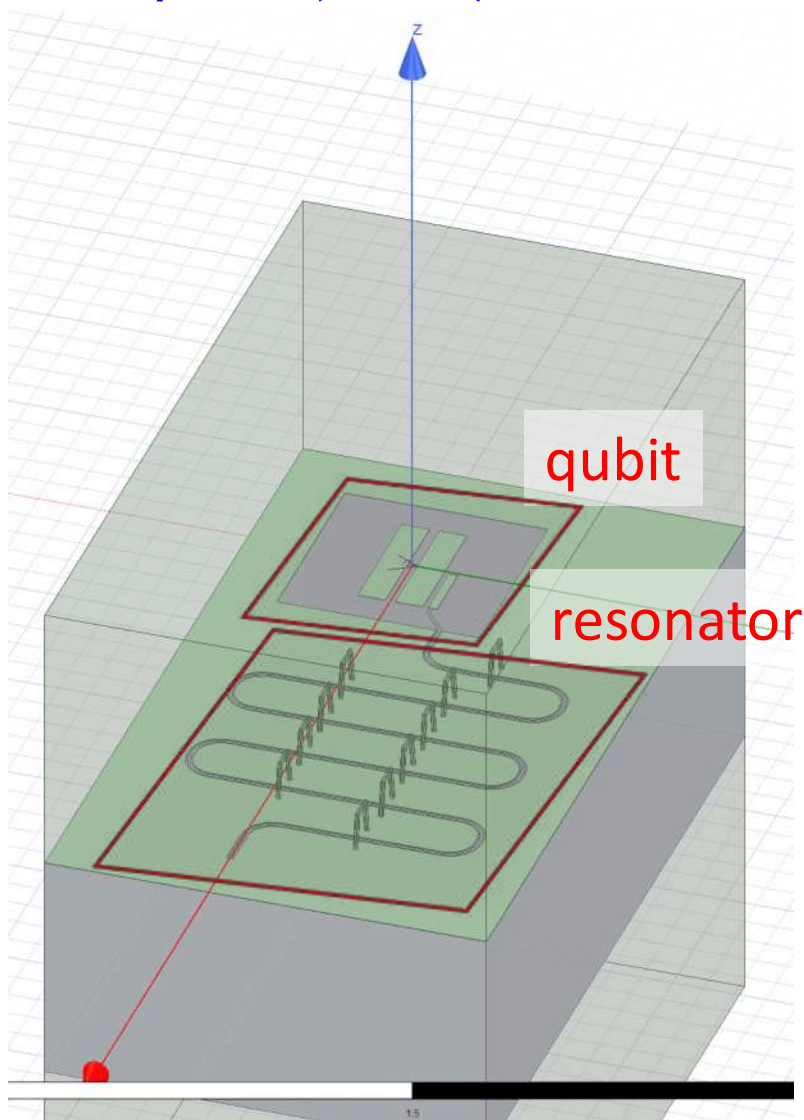
Circuit quantum electrodynamics (cQED) with modular quasi-lumped models

Zlatko K. Minev,* Thomas G. McConkey, Maika Takita, Antonio Corcoles, and Jay M. Gambetta
IBM Quantum, IBM T.J. Watson Research Center, Yorktown Heights, US

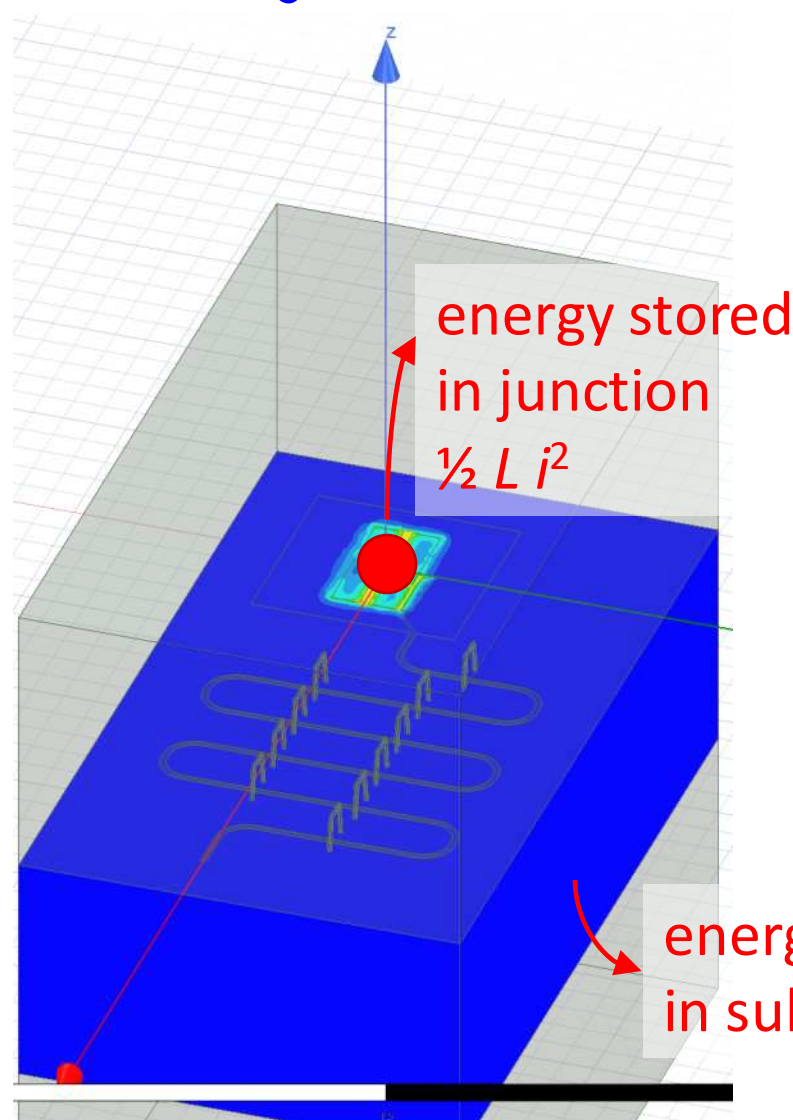
WIP: General capacitive analysis code
map to known building blocks: e.g.,
transmon, fluxonium, zero-pi, etc.

Example of step 2 with EPR

Physical (linear) model



Eigenmodes



Non-linear device
params

$$p_j \longrightarrow \hat{H}_{\text{full}}$$

Dissipation
budget

$$p_l \longrightarrow \mathcal{D}[\sqrt{\kappa}\hat{a}]$$

Step 3: Hamiltonian

Qubit/cavity anharmonicity

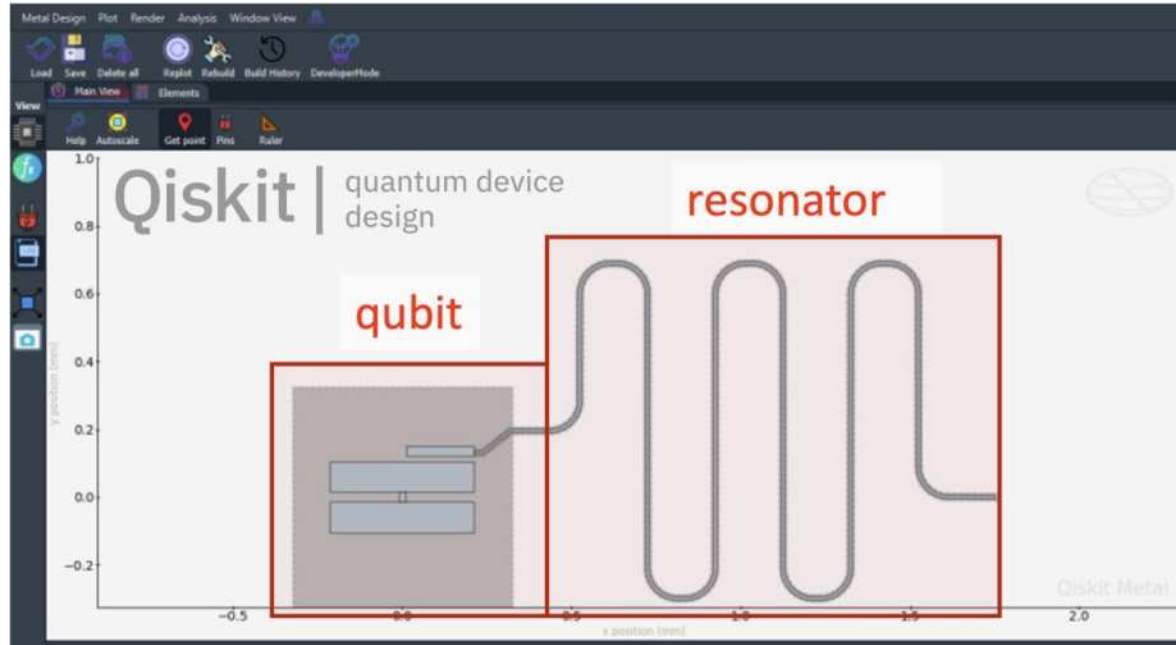
$$\alpha_{q/c} = p_{q/c}^2 \frac{\hbar \omega_{q/c}^2}{8E_J}$$

Qubit-cavity dispersive shift

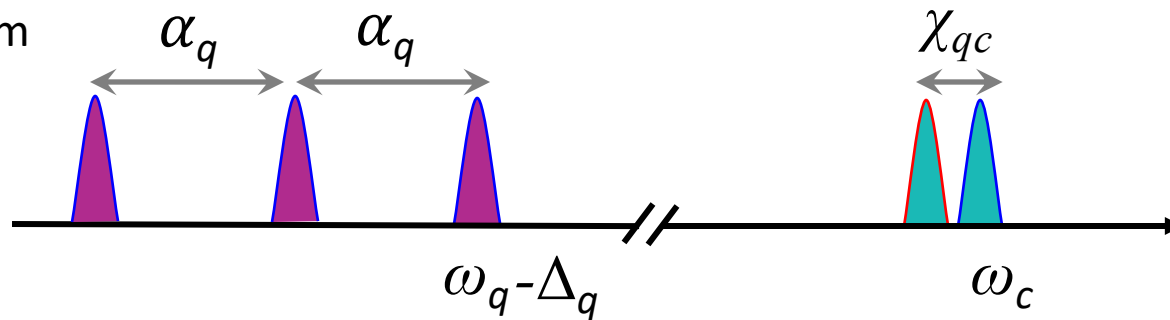
$$\chi_{qc} = p_q p_c \frac{\hbar \omega_q \omega_c}{4E_J}$$

Qubit Lamb shift

$$\Delta_q = \alpha_q - \frac{1}{2} \chi_{qc}$$



Transition spectrum



H_{eff} : for simplicity, showing up to $\mathcal{O}(\phi^6)$ in RWA

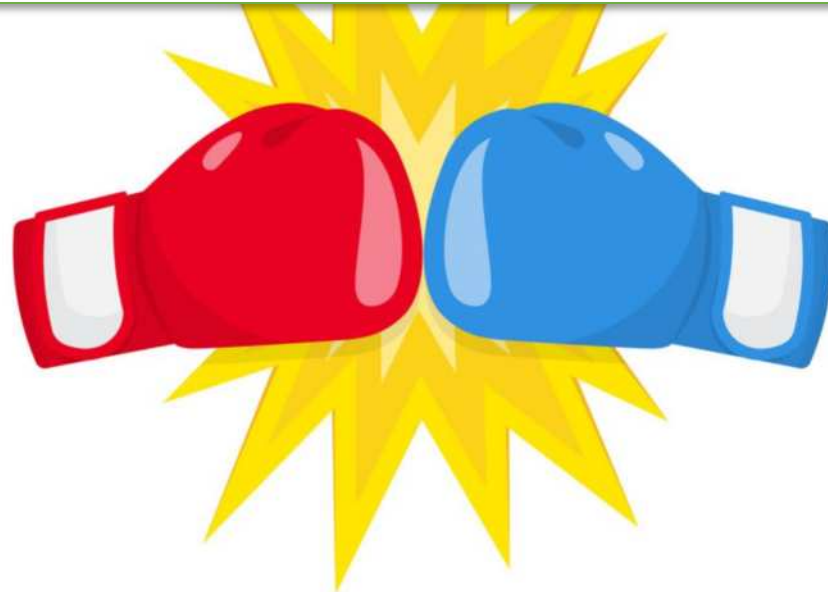
Planar devices & comparison to other methods

Quantum Physics

[Submitted on 2 Feb 2021]

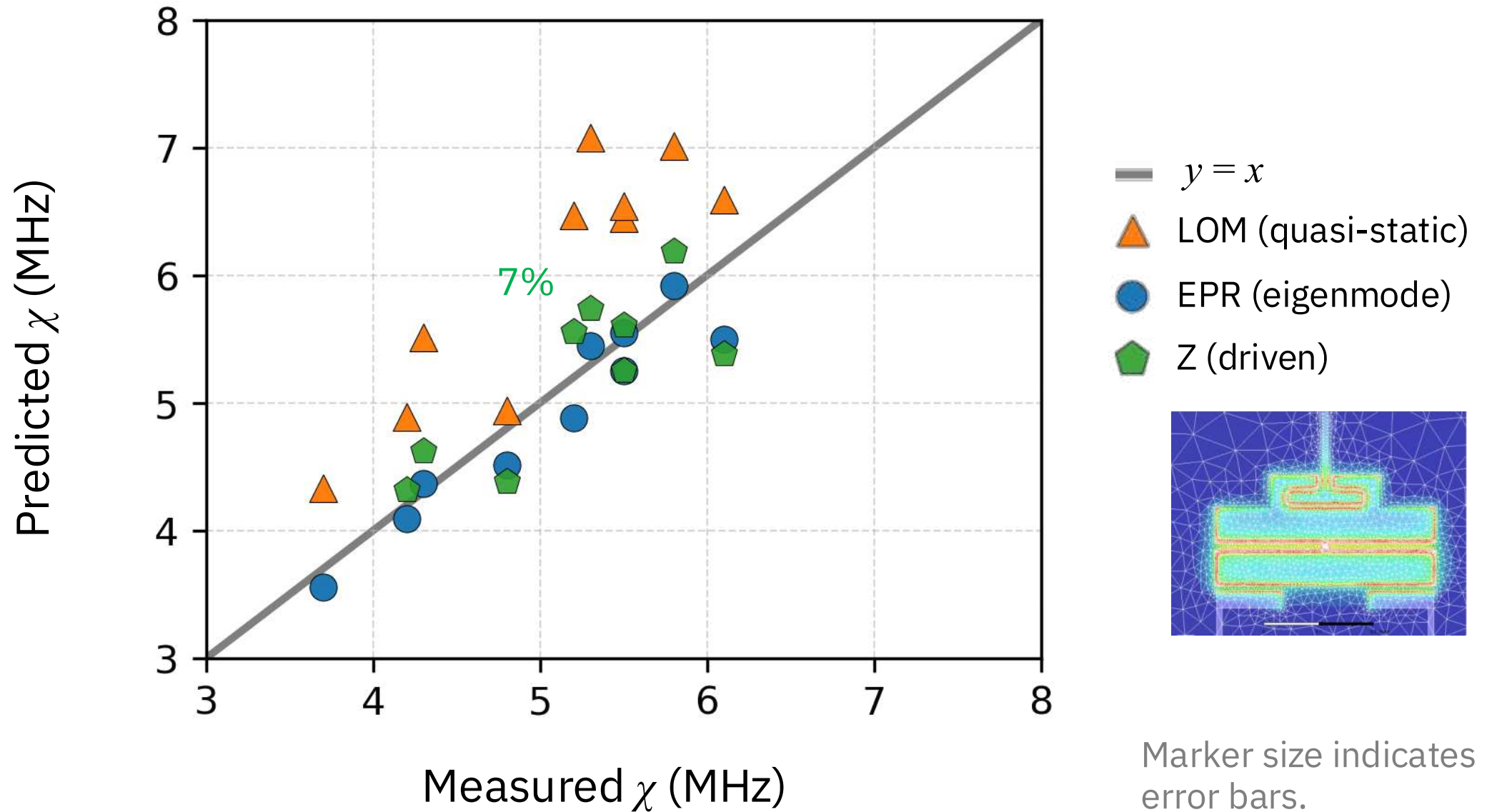
Exploiting dynamic quantum circuits in a quantum algorithm with superconducting qubits

Antonio D. Corcoles, Maika Takita, Ken Inoue, Scott Lekuch, Zlatko K. Minev, Jerry M. Chow, Jay M. Gambetta

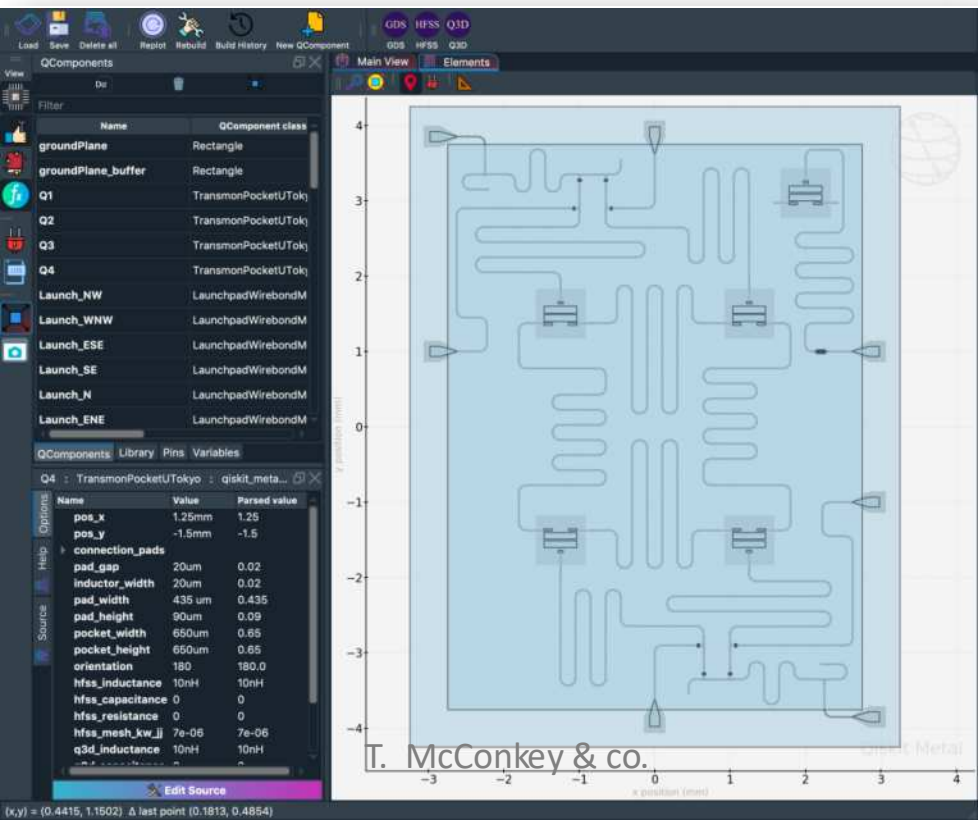


Automated with
Qiskit | quantum device
design

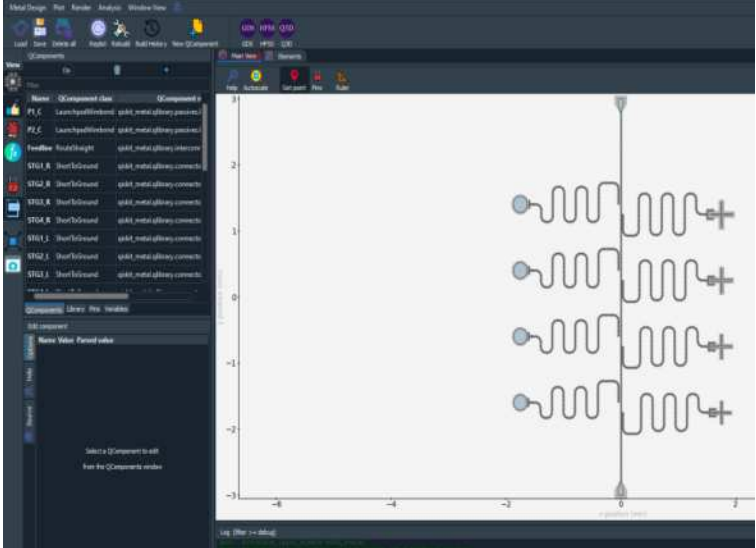
Measured vs. predicted: qubit-readout cross-Kerr



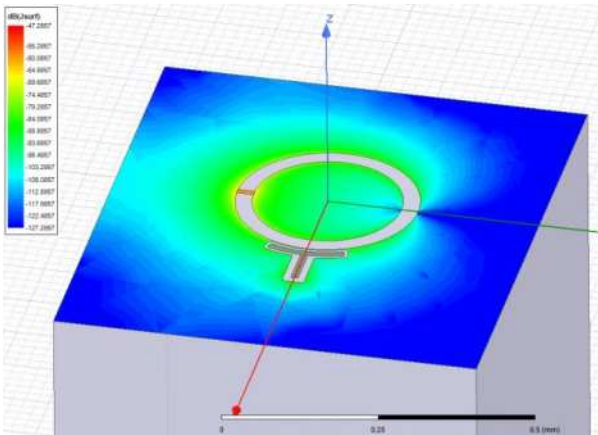
IBM 5Q Tsuru U Tokyo



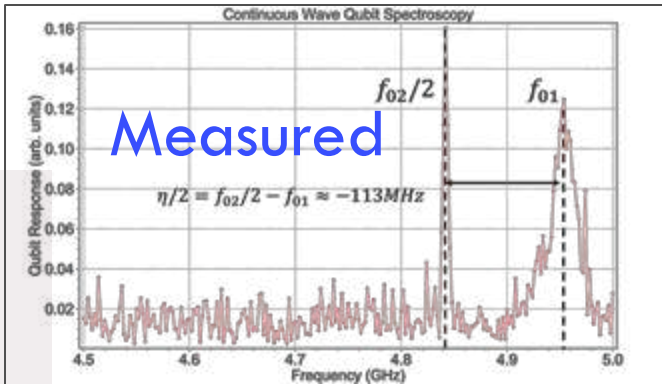
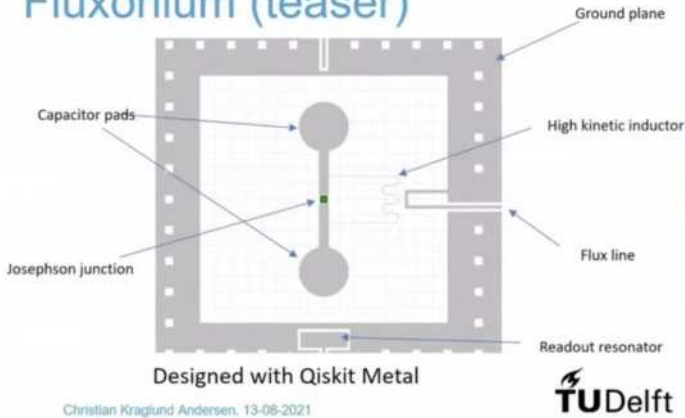
Chalmers 8Q



Simulated



TU Delft Fluxonium
Fluxonium (teaser)



Measured

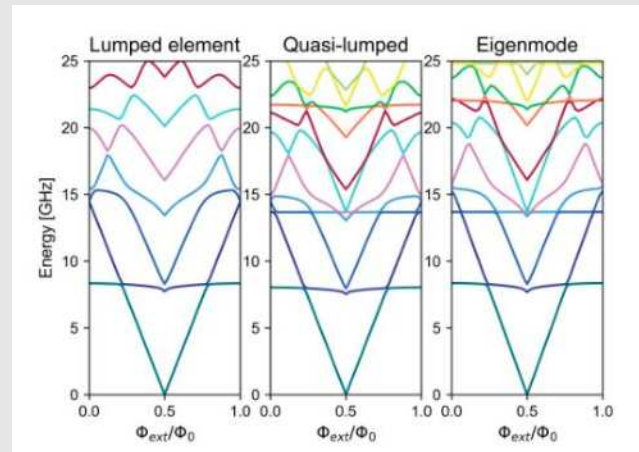
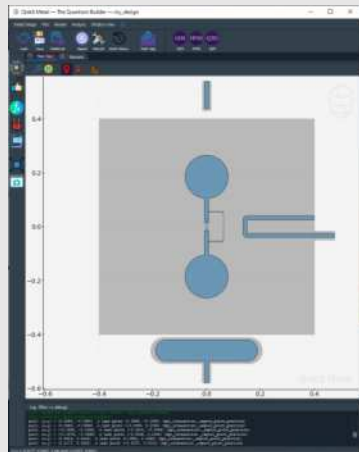


QuTech / Technische Universiteit Delft

Figen Yilmaz-Andersen Lab: Fluxonium chip design

Roald van den Boogaart-Andersen Lab: Fluxonium spectrum (eigenmode = pyEPR) to show how they have extended the analysis methods to be able to handle fluxonium qubits

Sara Buhktari-Andersen Lab: Fabricated resonator design being measured



3.1. Final design

The final design consists of an S-shaped feedline routed between two launchpads. Eight resonators are capacitively coupled to this feedline. Every component on this 9 x 9 mm device is implemented by a coplanar waveguide transmission line. The schematic representations of the full device and single resonator geometry are depicted in Figure 6.

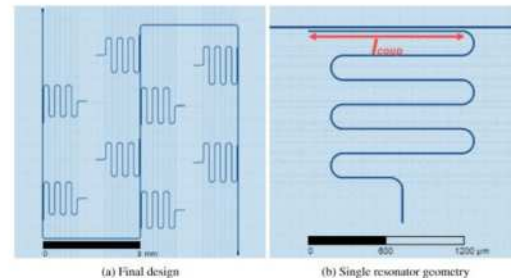
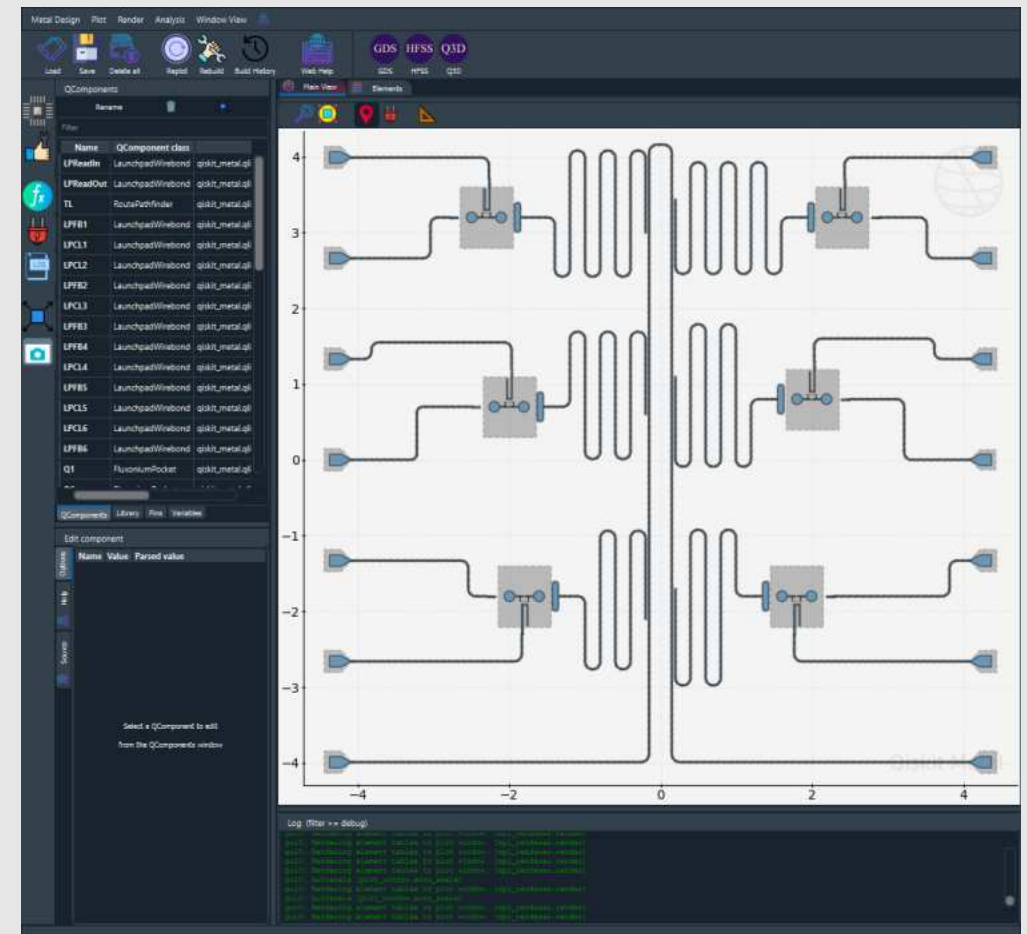


Figure 6: Schematic representations of the final design and single resonator geometry. Each resonator is capacitively coupled to the feedline. The resonator coupling length is denoted by l_{coup} .

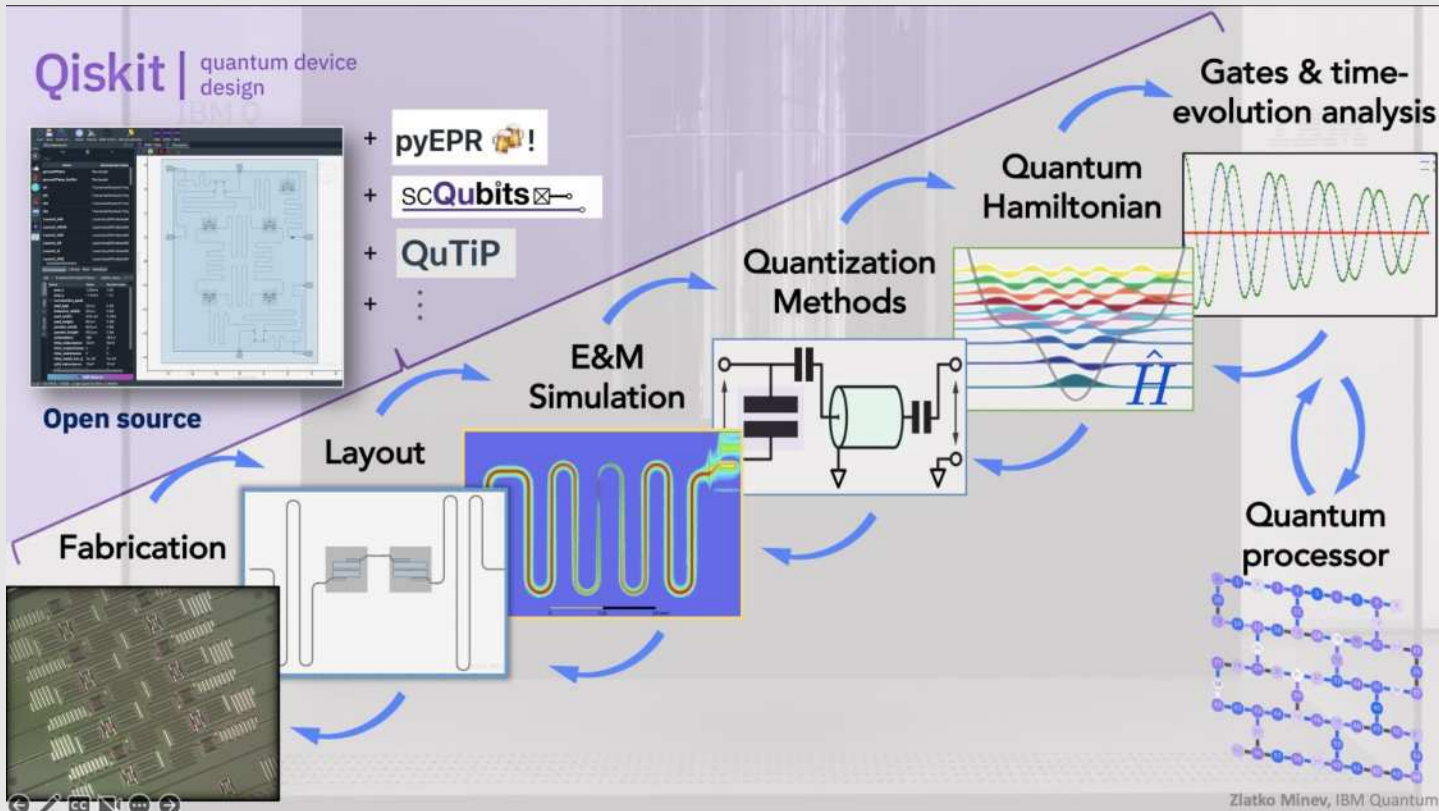
“Qiskit-Metal is the best tool if one studies superconducting qubits”
-- Figen Yilmaz

QuTech / Technische Universiteit Delft

Intermediate chip design.



Building it together



Open source

Call for community participation

Education

tutorials

courses?

summer schools with hands on (see QGSS 2020 Z. Minev from *Introduction to Quantum Computing and Quantum Hardware*)

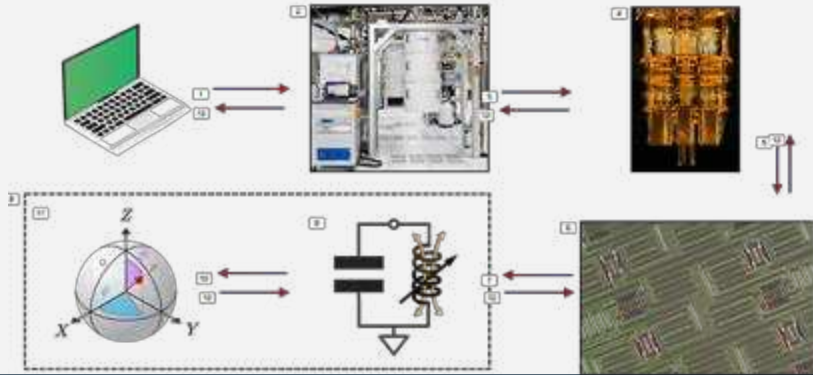
...



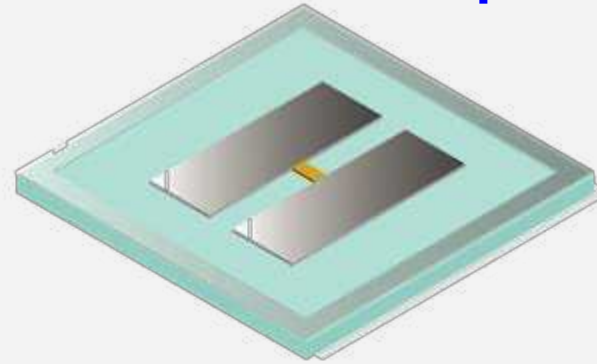
@zlatko_minev

The road behind and ahead

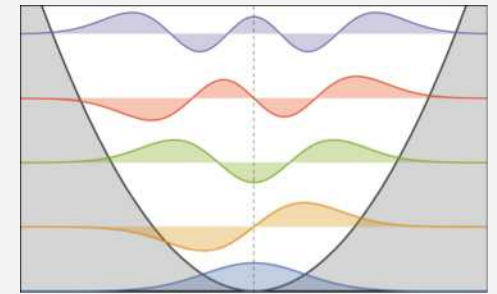
Qubit in the cloud



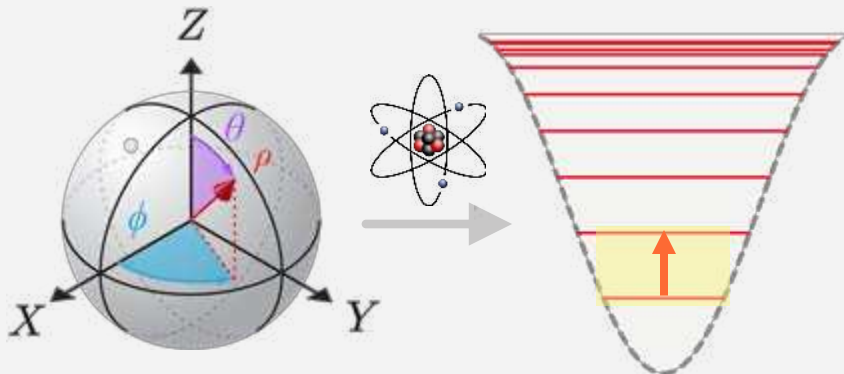
cQED: Transmon qubit



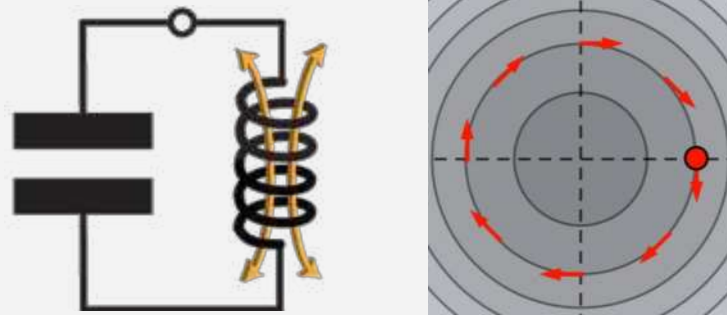
Unveiling the quantum oscillator



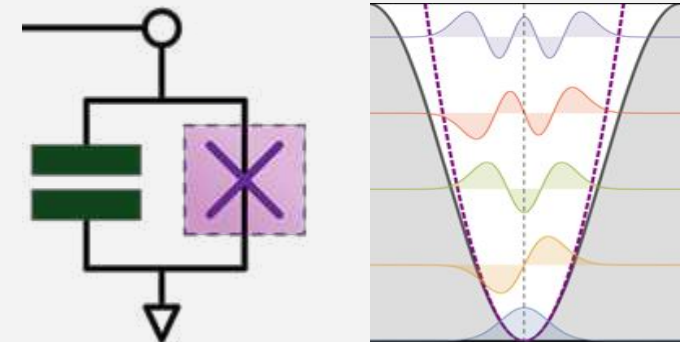
Qubit from atom / oscillator



Classical circuits & the LC



Transmon qubit



The important thing is not to stop questioning. Curiosity has its own reason for existence.

One cannot help but be in awe when he contemplates the mysteries of eternity, of life, of the marvelous structure of reality.

It is enough if one tries merely to comprehend a little of this mystery each day.

Albert Einstein

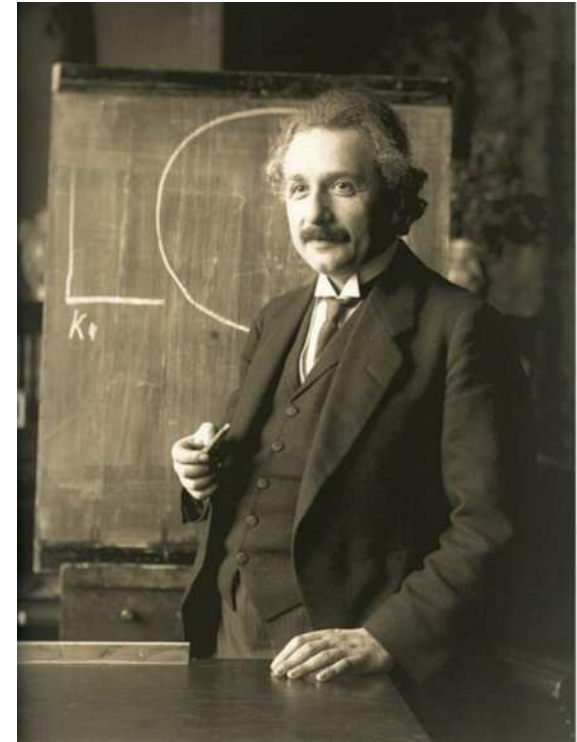


Photo: F. Schmutzer



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IBM Quantum

Next steps

Lab work with Dr. Murat Sarihan, Sadman Shanto, and more!

Check out references, problems given in the lecture, dangerous bends

Break away from the rules of today



Thank you!

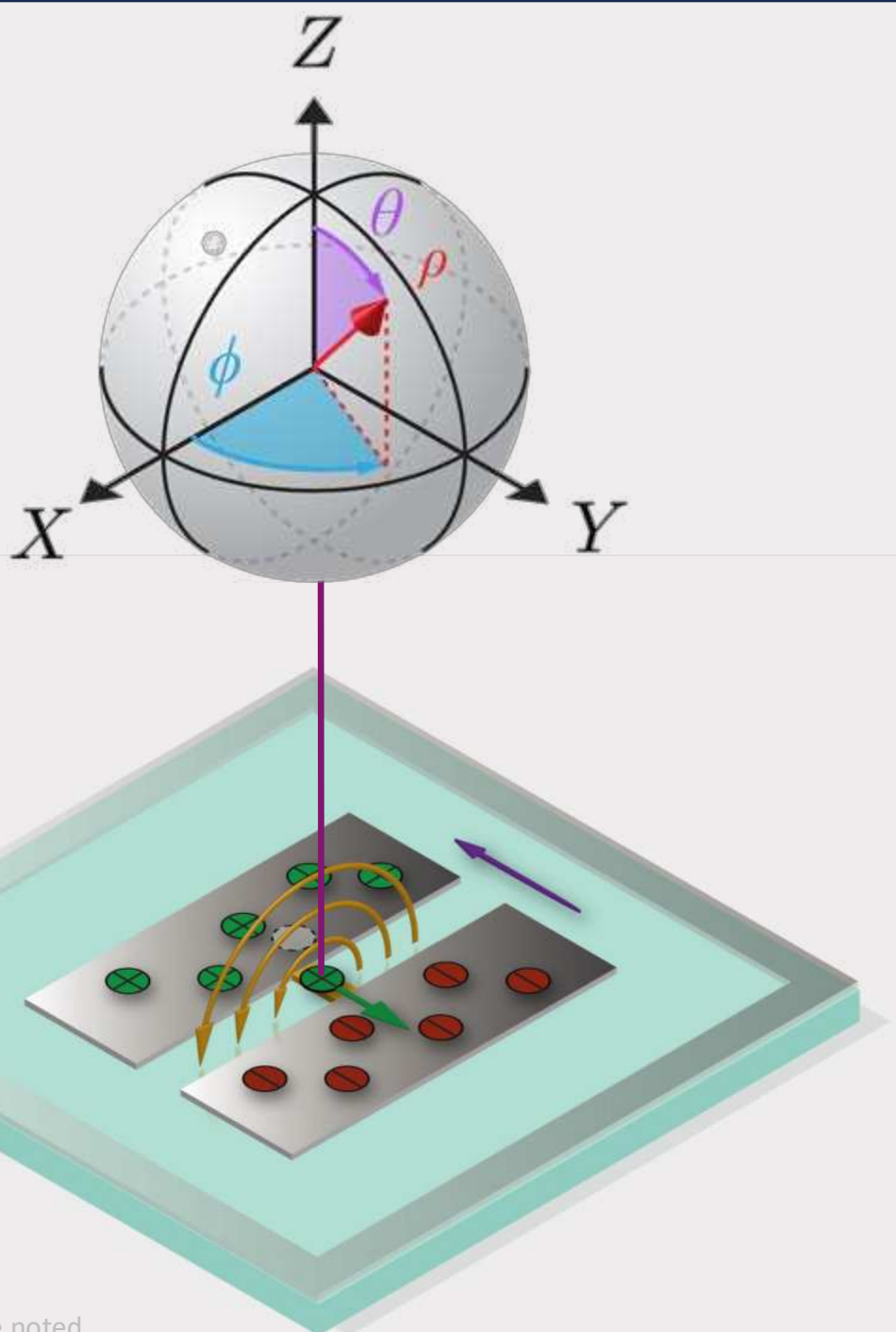
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Your First Superconducting Qubit

Introduction to **Circuit** Quantum Electrodynamics (cQED)

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Hand-written notes

Hamiltonian and energy

$$\begin{aligned}\hat{H} &= \frac{\hat{\Phi}^2}{2L} + \frac{\hat{Q}^2}{2C} \\ &= \hbar\omega_0 (\hat{a}^\dagger \hat{a} + \hat{a} \hat{a}^\dagger)\end{aligned}$$

Mean, variance, and RMS fluctuations

$$\langle 0 | \hat{\Phi}^2 | 0 \rangle = \Phi_{\text{ZPF}}^2$$

$$\langle 0 | \hat{Q}^2 | 0 \rangle = Q_{\text{ZPF}}^2$$

Calculations of the energy

For notational simplicity, I will drop the hats on the operators temporarily and the 0 from ω_0 .

$$\hat{H} = \frac{\hbar\omega}{2} (\hat{a}^\dagger \hat{a} + \hat{a} \hat{a}^\dagger)$$

$$\hat{a}^\dagger \hat{a} |n\rangle = n |n\rangle \quad \leftarrow \text{Fock}$$

$$\hat{a}^\dagger (\hat{a} |n\rangle)$$

$$\hat{a}^\dagger \sqrt{n} |n-1\rangle$$

$$\text{using } \hat{a}^\dagger |n\rangle = \sqrt{n+1} |n+1\rangle$$

$$\downarrow \sqrt{n+1-1} \sqrt{n} |n-1+1\rangle$$

$$\downarrow$$

$$= n |n\rangle$$

$\hat{a}^\dagger a = \text{photon number } n$

$$[a, a^\dagger] = 1$$

$$a a^\dagger - a^\dagger a = 1$$

$$a a^\dagger = 1 + a^\dagger a$$

$$\hat{H} = \frac{\hbar\omega}{2} (a^\dagger a + a a^\dagger)$$

$$= \frac{\hbar\omega}{2} (a^\dagger a + a^\dagger a + 1)$$

$$= \frac{\hbar\omega}{2} (2a^\dagger a + 1)$$

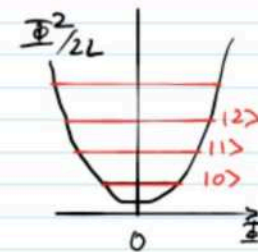
$$\hat{H} = \hbar\omega_0 \left(\frac{a^\dagger a}{n} + \frac{1}{2} \right)$$

n zero-point energy

$$\hat{H} |n\rangle = E_n |n\rangle$$

$$= \hbar\omega_0 (n + 1/2)$$

$$n = 0, 1, 2, \dots, \infty$$



Expectation value of magnetic flux and charge

$\langle \hat{\Phi} \rangle = \langle n | \Phi_{ZPF} (\hat{a} + \hat{a}^\dagger) | n \rangle$

$\leftarrow \text{some \#} = \sqrt{\frac{\hbar Z}{2}} \quad Z = \sqrt{L/C}$

$\langle n | \hat{a} | n \rangle = \sqrt{n} \langle n | n-1 \rangle = 0$

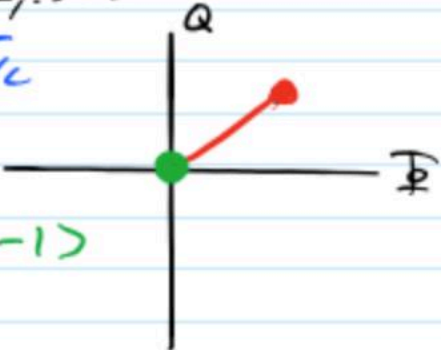
$\langle n | \hat{a}^\dagger | n \rangle = \sqrt{n+1} \langle n | n+1 \rangle = 0$

$\langle \hat{\Phi} \rangle = \Phi_{ZPF} (\langle \hat{a} \rangle + \langle \hat{a}^\dagger \rangle) = 0$

$\langle \hat{Q} \rangle = \langle n | -i Q_{ZPF} (\hat{a}^\dagger - \hat{a}) | n \rangle$

$= -i Q_{ZPF} (\langle \hat{a}^\dagger \rangle - \langle \hat{a} \rangle) = 0$

$Q_{ZPF} = \sqrt{\frac{\hbar}{2Z}}$



Fluctuations of flux

$$\text{Var}(\hat{\Phi}) = \langle \hat{\Phi}^2 \rangle - \langle \hat{\Phi} \rangle^2$$

$$= \langle \hat{\Phi}^2 \rangle$$

$$= \langle n | (a^\dagger + a)^2 | n \rangle \times \Phi_{\text{ZPF}}^2$$

$$= \langle n | a^{\dagger 2} + a^2 + a^\dagger a + a a^\dagger | n \rangle \times \dots$$

$$= \underbrace{\langle n | a^{\dagger 2} | n \rangle}_{\langle n | n+2 \rangle} + \underbrace{\langle n | a^2 | n \rangle}_{\rightarrow 0} + \langle a^\dagger a + a a^\dagger \rangle \times \Phi_{\text{ZPF}}^2$$

$$= \Phi_{\text{ZPF}}^2 \langle a^\dagger a + a a^\dagger \rangle$$

$$= \Phi_{\text{ZPF}}^2 \langle a^\dagger a + a^\dagger a + 1 \rangle$$

Hand-written notes

Day 2

$$\boxed{-E_5 \phi_{ZPF}^4} (\hat{a} + \hat{a}^\dagger)^4$$

Drop huts

$$(a + a^\dagger)(a + a^\dagger)(a + a^\dagger)(a + a^\dagger)$$

$$aaaa + aaaa^\dagger + aa^\dagger aa + \dots$$

$$aa^\dagger aa$$

↳ wave mixing

$$aa^\dagger aa$$

$$\{a, a^\dagger\} = 1$$

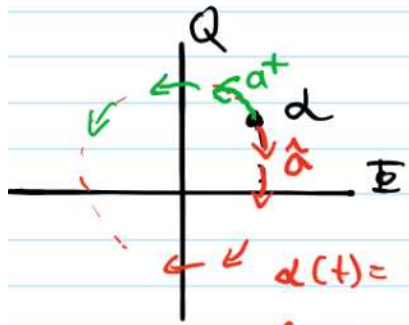
$$aa^\dagger - a^\dagger a = 1$$

$$aa^\dagger = 1 + a^\dagger a$$

$$(a^\dagger a + 1)aa$$

$$a^\dagger a aa + aa$$

Hand-written notes

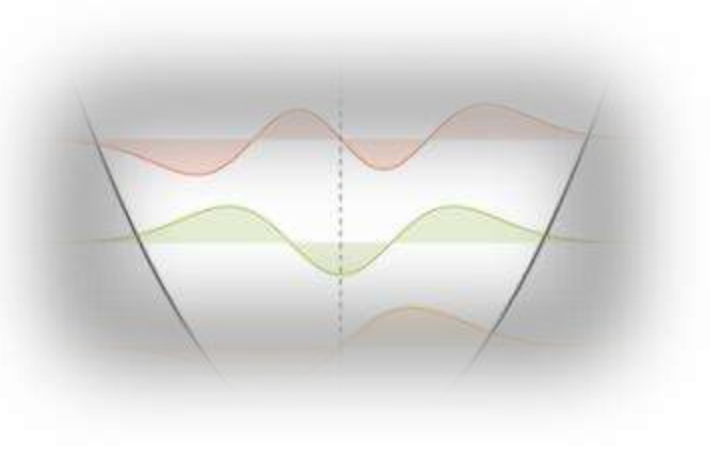


$a^\dagger a^3 + a^2$
 $\uparrow$ 4-wave mixing
 $\nwarrow$ non-classical term

$\left. \begin{aligned} d(t) &= d(0) e^{-i\omega_0 t} \\ \hat{a}(t) &= \hat{a}(0) e^{-i\omega_0 t} \end{aligned} \right\}$
 $\hat{a}^\dagger(t) = \hat{a}^\dagger(0) e^{+i\omega_0 t}$

$\hookrightarrow a^\dagger a^3 = a^\dagger e^{+i\omega_0 t} a^3 e^{-3i\omega_0 t}$
 $= a^\dagger a^3 e^{-2i\omega_0 t}$ rotating
 $\hat{a}^{\dagger 2} \hat{a}^2 = (\hat{a}^\dagger e^{+i\omega_0 t})^2 (\hat{a} e^{-i\omega_0 t})^2$
 $= (\hat{a}^\dagger)^2 (\hat{a})^2 e^{+2i\omega_0 t} e^{-2i\omega_0 t}$
 $= \hat{a}^{\dagger 2} \hat{a}^2$ non-rotating

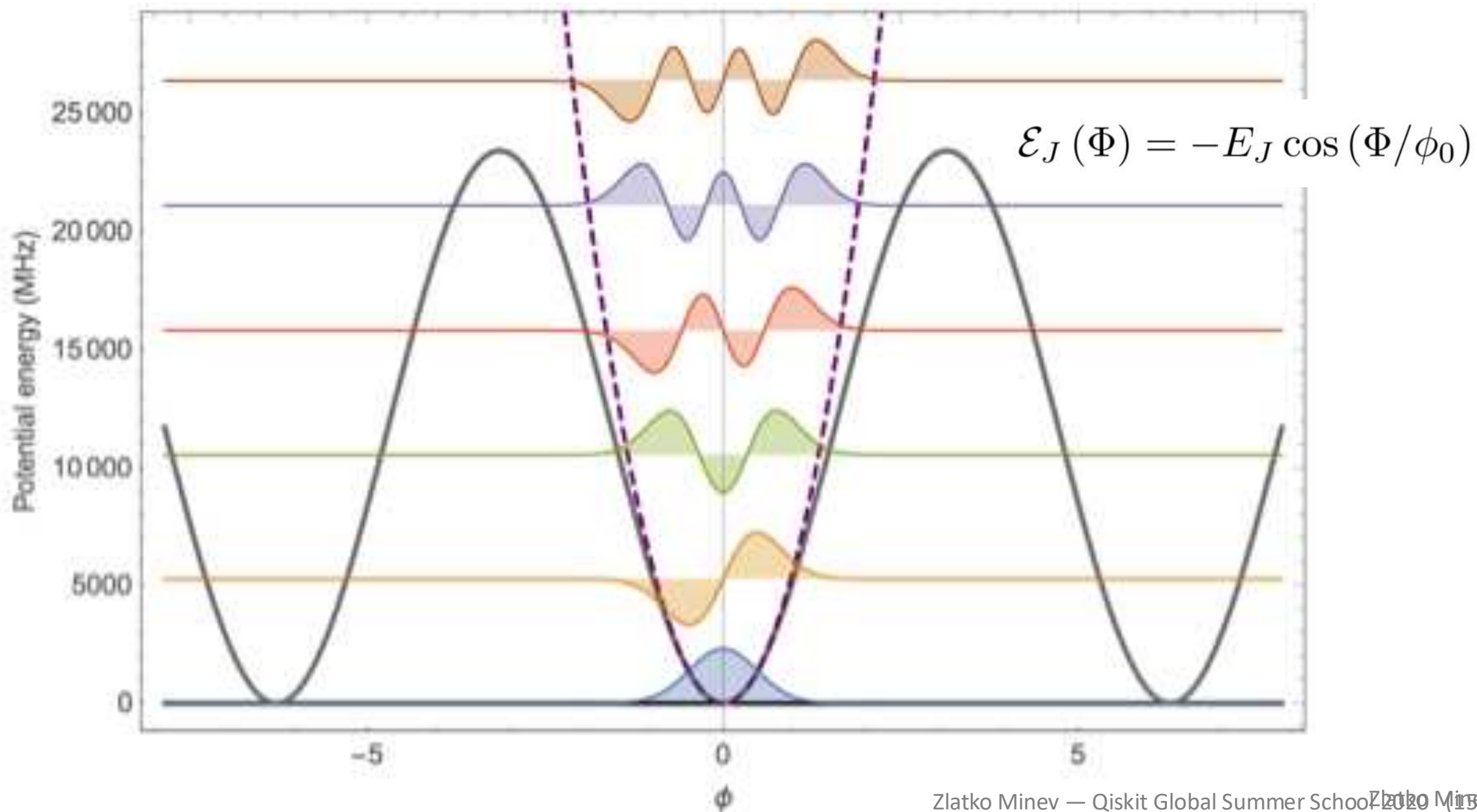
$\hat{H}_0 = \hat{H}_0 + \hat{H}_R(t)$
 $|^{st}_{app}, \hat{H} \approx \hat{H}_0$



Is the junction phase/flux compact or not?



Harmonic Approximation



Tight-binding model

